

Finite Separators and Discriminant Factorizations for Sextic Block Resolvents

A sharp 196-point guarantee and a bounded replacement
for ProveIt's separating-invariant search

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ProveIt research-report collection, continuing the formal project Algebra/PolynomialFormulas. Batches 40–41 intake, batch 41, manuscript 03 (archive ProveIt_Finite_Sextic_Separators); pinned to ProveIt e2b1f016a. Part II: batch 57, manuscript 03 (archive ProveIt_Three_Plus_One_Resolvents), pinned to e47ed8547. AI-assisted, unrefereed, not formalized. Notes marked **[write]** were added when the manuscripts were written into the collection; see [sections 1.4, 1.5](#) and [18](#).

Abstract

The sextic radical-solvability development in ProveIt uses terminating but unbounded search to specialize two block-partition descriptors. We replace that search, at the level of an ordinary mathematical proof, by a single explicit parameter range. For every separable sextic in characteristic zero, one of the integers $1, \dots, 196$ separates all fifteen pair partitions and all ten triple partitions simultaneously. The underlying descriptors compress to quadratics and linears. Their collisions are controlled by three invariant polynomials Q_f, U_f, L_f , with degrees at most 30, 120, 45, and we prove

$$\text{disc}_Z R_{2,f} = \text{disc}(f)^6 Q_f^6 U_f^2, \quad \text{disc}_Z R_{3,f} = \text{disc}(f)^3 L_f^2.$$

For the generic sextic, the three factors are irreducible and have the stated exact degrees. Consequently 196 is optimal for a guarantee valid for every set of distinct nonzero complex test parameters, although optimality of the initial integer interval is not established. A split integer example gives a lower bound of eight for that latter problem. We derive a coefficient-only primitive-recursive decision architecture, a height-only separator requiring no search, a precise repository integration plan, and further research questions. Exact symbolic checks and a modular Bézout certificate accompany the proofs.

Status and scope. This is an unrefereed research article with complete written proofs and reproducible exact checks. The new results have not been formalized or run in Lean. General decidability, including polynomial-time decidability of solvability by radicals, is classical; it is not claimed as a new discovery. The specific separator and collision-factorization results are proposed contributions relative to the inspected repository. Historical priority is not certified by the literature check.

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1 The repository question and the contribution

The starting point is a concrete implementation boundary, not an assertion that sextic solvability is an unsolved problem. In the pinned ProveIt snapshot, `SexticRadicalDecision.lean` proves a total recursive criterion for integer sextics. Its introductory comment explicitly does not claim primitive recursiveness, because the separating-invariant search uses unbounded minimization. The supporting file `SexticIrreducibleDecision.lean` constructs pair and triple parameters using `Nat.find`. The file `SexticSeparatingInvariants.lean` proves that suitable parameters exist by representing each block by its root polynomial and each partition by the polynomial of those block polynomials [4, 5, 6, 7].

That is a particularly useful research target: the existing semantic architecture already identifies the right invariant and its correctness condition. What is missing from this route is a small explicit bound and a sufficiently transparent explanation of the exceptional specializations. Our answer is stronger than merely bounding an enumeration of pairs of natural numbers. A one-dimensional curve works for every distinct root tuple, its failure set has three explicit components, and its generic number of failures can be determined exactly.

1.1 What is classical, inherited, and proved here

The equivalence between radical solvability and solvability of the Galois group is classical. So is the sextic criterion using the two imprimitive wreath-product groups of degrees fifteen and ten in the corresponding coset actions; see, for example, Boswell and Glasser [2]. Landau and Miller proved polynomial-time decidability of radical solvability in 1985 [1]. Thus even a primitive-recursive decision for sextics is not, as an abstract existence theorem, new.

The partition descriptors and the strategy of separating their specialized values are inherited from the inspected ProveIt source. More broadly, the construction and use of relative invariants are central to computational Galois theory; Fieker and Klüners discuss general methods and the practical importance of structured invariants [3].

The claims established in this article are the explicit compression, the special curve $(u, v) = (t, 2t^2)$, the 196-point uniform bound, the three-factor description of its obstruction, the exact discriminant identities, and generic irreducibility and sharpness. Their proofs are given below rather than inferred from computations. The bounded decision architecture is an application to the repository’s mathematical gap, not a replacement claim for the stronger classical complexity theorem.

The literature check consulted the primary sources just cited and the pinned source files. It does not establish that no equivalent identity has appeared elsewhere under different coordinates or resolvent conventions. Accordingly, “contribution” here means a result developed and proved for this report, with a demonstrated use relative to this repository; it does not mean a certified first-in-literature result.

1.2 Main results at a glance

The root tuple is always assumed distinct. The polynomials h_P and k_T are defined explicitly in [section 3](#); their resolvents have degrees fifteen and ten in Z .

Uniform separator. Among any 196 distinct nonzero elements of a characteristic-zero field containing the roots, there is a parameter t for which both maps $P \mapsto h_P(t)$ and $T \mapsto k_T(t)$ are injective. In particular, testing $1, \dots, 196$ always succeeds.

Exact obstruction. There is a nonzero polynomial $E_f = Q_f U_f L_f$ of degree at most 195, with coefficients polynomially expressible in the sextic coefficients, such that $E_f(t) \neq 0$ is exactly simultaneous separation. Its three factors have generic degrees 30, 120, 45 and are generically irreducible.

Sharpness and implementation. The number 196 is optimal for arbitrary complex test sets. For the initial integer interval the proven bounds are $8 \leq N_{\text{int}} \leq 196$. The explicit finite range removes unbounded minimization from a coefficient-only sextic decision architecture.

[write] Scope of the last sentence. The finite range removes unbounded minimization from the architecture of [section 11](#), on paper. It removes nothing from the Lean development yet: none of this report is formalized, and the Lean search it bounds is still an unbounded `Nat.find`; see [section 1.5](#).

1.3 [write] Conventions and notation

No symbol of the delivered text was renamed and no normalization was changed. The table lists the letters that carry more than one meaning in the text, in the Lean development it continues, or in the sibling report [specialization-safe-radical-solvers \[13\]](#); where a false reading is tempting it is printed beside the true one.

Symbol	Meaning here, and readings to avoid
t	The one parameter on the curve $(u, v) = (t, 2t^2)$ in descriptor space. Lean's parameter is a pair $x: \text{Fin } 2 \rightarrow \mathbb{N}$, with X0 the outer (block) variable v and X1 the root variable u . <i>True:</i> $x(0) = 2t^2, x(1) = t$. <i>False:</i> $x(0) = t, x(1) = 2t^2$.
n (Lean)	The <code>Nat.find</code> index of the Lean search is a <code>Primcodable</code> code, decoded to a pair by <code>parameterAt</code> ; it is not t . The bound 196 bounds t , not the code.
$p_B(u), p_b, p$	p_B is the root polynomial of a block B of any size, equation (2.3) ; $p_b = \alpha_i \alpha_j$ is the product of a root pair b , equation (3.1) ; $p = 1,000,003$ is the prime of equation (9.3) .
$e_j, e_j(B), e$	e_j are the elementary symmetric functions of all six roots, $e_j(B)$ those of a block; e, e' are root pairs. In lemma 5.1 the letters $a, \dots, f$ are root values, so there f is not the sextic and e is not a pair.
$A_P, B_P, C_P; a_T, b_T; A, B$	Pair and triple coefficients, equations (3.1) and (3.4) . The bare A, B are the coefficients of $h_P - h_Q$ as a linear polynomial in one root in the proof of lemma 8.2 , and the Bézout cofactors in equation (9.3) .
$R_{2,f}, R_{3,f}, R$	The compressed resolvents equation (7.1) , of degrees 15 and 10 in Z , are built from the values $h_P(t), k_T(t)$; Lean's evaluated resolvents are built from the descriptor values $D_P(t, 2t^2) = K_2(t) - 2t^2 h_P(t)$, an affine image, so the two are separable at the same $t \neq 0$. $R = 1 + H$ is a root bound (section 11). Neither is the sibling report's subset-sum resolvent $R_{p,k,f}$.
$h_P(t)$	The pair invariant equation (3.1) , quadratic in t ; not the sibling report's subset compression $h_A(u) = \sum_r u^{r-1} e_r(x_A)$.
K, K_2, K_3	K is the field; K_2, K_3 are the partition-independent parts in equations (3.3) and (3.6) .
H, H_R	H is the largest absolute value of the lower coefficients of the monic sextic (sections 11 and 12); in the proof of lemma 8.2 H is a putative common factor; H_R is the analogue of H for a resolvent.
Q, U, L, E and Q_f, U_f, L_f, E_f	Universal polynomials over $\mathbb{Q}[e_1, \dots, e_6, t]$ (section 8) and their specializations to one sextic; $\bar{E}_f$ is E_f reduced modulo p .
Degree	Unqualified degrees 30, 120, 45, 195 are degrees in t . The homogeneous degrees 30, 240, 90, 360 of the remark after proposition 6.2 count roots and t together.

Symbol	Meaning here, and readings to avoid
196 versus N_{int}	196 is sharp for <i>arbitrary</i> sets of distinct nonzero complex test values (theorem 9.2). For the initial integer interval only $8 \leq N_{\text{int}} \leq 196$ is proved (proposition 9.3). <i>False reading</i> : “some sextic defeats all of $1, \dots, 195$ ”.

1.4 [write] Provenance of this report

This report is built from **one** manuscript: batch 41, manuscript 03 of ProveIt’s incoming-reports intake (written together with batch 40), delivered as the archive `ProveIt_Finite_Sextic_Separators` (11 files, including a 24-page Letter PDF) in commit `9754e8360`, pinned to ProveIt commit `e2b1f016a94102663f12b970e6dd434229f90d01`, and placed in commit `eaf787d50`. It is filed in ProveIt’s research-report collection beside `specialization-safe-radical-solvers` [[13](#)], with which it shares no theorem, and continues the formal project `Algebra/PolynomialFormulas`. That project is unchanged between the pin and the placement commit, so every repository statement of the delivered text was checked against the tree it describes. Every statement, proof, table, example, remark, question and limitation of the delivered text is printed. No merge choice arose.

The write changed only the following. Every label carries the prefix `sbr:`. The title-page provenance line, [sections 1.3 to 1.6](#), the unnumbered **[write]** notes, and the bibliography entries [[11](#)], [[12](#)] and [[13](#)] were added, and the title page no longer duplicates the PDF anchor of page 1. Two corrections of form: the title-page spacing was tightened so that the added line fits, and the environments sharing the theorem counter now use alias counters, because in the delivered build every cross-reference to a lemma, proposition or corollary printed the word “theorem”. The notes use no theorem counter, so every theorem, equation and section number is that of the delivered PDF. The delivered text names the delivery layout: its files `checks/X` are shipped as `data/X`, its `build.sh` as `code/build.sh` and its `requirements.txt` as `data/requirements.txt`; the delivered PDF, README and checksum file `SHA256SUMS` are not shipped (the checksums were verified, 10 of 10, at placement). [Section A](#) carries a note with commands that work with the shipped names.

Part II. On 29 September 2026 a second manuscript, batch 57 manuscript 03 (*Three Plus One: Radical-Solvability Tests Without Separating Sextic Resolvents*, archive `ProveIt_Three_Plus_One_Resolvents`, pinned to ProveIt commit `e47ed8547bd021f9bd85bdcf767c94a2fcb41001`, placed in `70b39943a`), was added as Part II; its provenance, notation and status are in [section 18](#), and its files carry the prefix `02-three-plus-one-`. It was written against the text of Part I printed here. In Part I, “the manuscript” and “the delivered text” always mean the batch-41 source. Part I is unchanged except for the title-page line, this paragraph, a contents line, labels on three research questions, dated notes after [theorem 10.3](#) and those questions, and bibliography annotations. Neither part is formalized.

Relation to the sibling report. The manuscript does not know `specialization-safe-radical-solvers` [[13](#)]. That report’s separating repair compresses the k -subsets of the roots of a prime-degree polynomial by $h_A(u)$ and bounds the bad parameters by adding the degrees of all pairwise differences (1191 values suffice for septic triples, not claimed sharp); its question on smaller deterministic separating families asks for a better number for $(n, k) = (7, 3)$ by counting collision orbits under the Galois action. The present report carries out that kind of orbit count for a different family, the sextic block partitions, and reaches a sharp bound. It answers no question of the sibling report.

1.5 [write] Formal status in ProveIt

None of this report’s statements is formalized. No `sbr` : label has a Lean or Rocq counterpart. In particular the compression ([propositions 3.1](#) and [3.2](#)), the collision factorizations, the exact failure test [equation \(6.6\)](#), the 196-point bound ([theorem 9.1](#)), the discriminant identities, generic irreducibility, sharpness, the N_{int} bracket, the bounded architecture ([theorem 11.1](#)) and the height separator have written proofs only. The modules proposed in [section 14](#) (`SexticCompressedDescriptors.lean`, `SexticCollisionFactors.lean`, `SexticBoundedSeparatingSearch.lean`) do not exist. Placement beside a Lean and Rocq development confers no formal status.

What ProveIt does prove in Lean. All in namespace `LeanProofs.PolynomialFormulas` under `Algebra/PolynomialFormulas/Lean/PolynomialFormulas/`, at the pin and unchanged since; line numbers are those of the declarations.

- `SexticSeparatingInvariants.lean` [7]: `rootPolynomial_injective` (l. 115), `pairDescriptor_injective` and `tripleDescriptor_injective` (l. 422, 428), which together are [lemma 2.1](#); `bivariateDescriptor` (l. 618), a product of X_0 minus block polynomials in X_1 , and `pairBivariateDescriptor_injective`, `tripleBivariateDescriptor_injective` (l. 668, 677); `pairDescriptorValue`, `tripleDescriptorValue` (l. 790, 794), the evaluation at x : $\text{Fin } 2 \rightarrow \mathbb{N}$; `exists_nat_eval_injective` (l. 1048), which separates a finite injective family of multivariate polynomials over a characteristic-zero domain by one natural-number point (through `exists_nat_eval_ne_zero`, l. 1019, the combinatorial Nullstellensatz); and its sextic instances `exists_pairDescriptor_separating_evaluation`, `exists_tripleDescriptor_separating_evaluation` (l. 1074, 1083) and `exists_pairDescriptorValue_injective`, `exists_tripleDescriptorValue_injective` (l. 1092, 1102). These statements are purely existential. The Nullstellensatz proof finds the point in a finite grid, but no bound is stated or exported, and nothing restricts the point to a curve.
- `SexticSeparatingSearch.lean` [11]: the Boolean tests `pairSeparatesB`, `tripleSeparatesB` (l. 536, 539), through the collision products of `pairCollisionPolynomial` (l. 114) and `collisionProduct` (l. 318); the unbounded searches `pairSeparatingCode`, `tripleSeparatingCode` (l. 604, 608; `Nat.find` at l. 606, 610) over codes decoded by `parameterAt` (l. 517); the parameters `pairSeparatingParameter`, `tripleSeparatingParameter` (l. 612, 616); and their correctness `pairSeparatingParameter_injective`, `tripleSeparatingParameter_injective` (l. 620, 629). This is the search this report bounds. *The manuscript does not cite this file*: its citation [6] is `SexticIrreducibleDecision.lean`.
- `SexticIrreducibleDecision.lean` [6]: the guarded total searches `pairTotalCode`, `tripleTotalCode` (l. 87, 90; `Nat.find` at l. 88, 91), total on every monic sextic because a bounded reducibility test returns code 0 on reducible inputs, and `pairTotalCode_computable` (l. 101).
- `SexticRadicalDecision.lean` [5]: the public results `sexticRadicalDecision_correct` (l. 34), `allRootsRadical_computablePred` (l. 49) and `has_verified_sextic_radical_turing_machine` (l. 56). Its docstring (l. 8–9) calls the criterion “not claimed primitive recursive, because the separating-invariant search uses unbounded minimization”.

The Rocq route (`Algebra/PolynomialFormulas/Coq/SexticMuRec*.v`) also selects the separating parameters “by genuine unbounded minimization”, as the project README states; the manuscript mentions it only in passing and does not analyse it.

What formalizing the bound would give (a formalization target, not a claim). A Lean proof of [theorem 9.1](#) for the root tuple of an irreducible monic sextic, together with the affine relations [equations \(3.3\)](#) and [\(3.6\)](#) and the convention $x = (2t^2, t)$ of the first table row, would make the Lean search provably bounded: by minimality of `Nat.find`, `pairSeparatingCode` would be at most the largest `encodeParameter` of the points $(2t^2, t)$, $1 \leq t \leq 151$, and `tripleSeparatingCode` the same with $t \leq 46$; one common

$t \leq 196$ serves both. Equivalently the search could be replaced by a fixed scan of $t = 1, \dots, 196$. That bounds the parameter selector only. The Lean pipeline has two further unbounded searches: `Sextic ReducibleDecision.linearTotalCode` (l. 72, `Nat.find` at l. 73), which the manuscript notes in [section 14](#); and `SexticSparseSymmetricSearch.elementaryCode` (l. 634, `Nat.find` at l. 637) [12], the certificate search behind `elementarySparse` (l. 671), which produces the elementary-symmetric formulas `pairElementarySparse`, `tripleElementarySparse` (l. 1133, 1138) for the resolvent coefficients and `pairCollisionElementary`, `tripleCollisionElementary` (`SexticSeparatingSearch.lean`, l. 282, 286) for the collision products. The manuscript does not mention this search. Its inputs are the parameter and a coefficient index, not the sextic, so over a finite parameter range it searches for finitely many fixed certificates; a `Primrec` proof must still bound or replace it. This is the condition, in [section 14](#), that the coefficient evaluators themselves have primitive-recursive proofs.

1.6 [write] What is not claimed

The delivered text states each of the following, and each is kept where the text states it. The article is unrefereed. Its new results were not formalized or run in Lean or Rocq, the repository was not rebuilt or modified, and nothing was benchmarked against production Galois software. Decidability of radical solvability, even in polynomial time, is classical (Landau and Miller), as is the sextic block criterion; the contributions are relative to the inspected repository and the consulted sources, and no exhaustive priority search was made. The Python files are an exact verifier and a root-level prototype, not a coefficient-only solver and not an extraction of a formal proof; the expanded universal coefficient tables are specified but not supplied. The value 196 is sharp for arbitrary complex test sets only; the initial integer interval is bracketed, not determined. The fixed-parameter counterexamples are reducible and say nothing about irreducible rational sextics. [Theorem 11.1](#) is a mathematical realization, not a changed Lean function or an accepted `Primrec` proof, and it claims no polynomial bit complexity. The height separator is a termination certificate or comparison construction, not a recommendation for practical evaluation. The Kronecker-curve principle ([proposition 15.1](#)) is a standard monomial-encoding argument, not a novelty claim.

At the write the two scripts were rerun on a copy with SymPy 1.14.0; both recorded outputs were reproduced exactly, apart from line endings. The write also checked, by a separate script, the Lean convention of the first table row against the definitions (300 random integer root tuples, each at $t = 1, \dots, 11$, no disagreement between Lean-style descriptor values and h_P, k_T), the first separator 8 of $(-3, 0, 1, 2, 3, 4)$, and, over $\mathbb{Z}$, that E_f for the roots [equation \(9.1\)](#) has degree 195, a nonzero constant term and no repeated factor. None of this is an independent proof review.

2 Notation and complete partition descriptors

Let K be a field of characteristic zero, let $\alpha_1, \dots, \alpha_6 \in K$ be distinct, and put

$$f(X) = \prod_{i=1}^6 (X - \alpha_i) = X^6 - e_1 X^5 + e_2 X^4 - e_3 X^3 + e_4 X^2 - e_5 X + e_6. \quad (2.1)$$

If the coefficients originally lie in a smaller field, take K to contain a splitting field. Our arguments are algebraic and do not depend on an ordering, numerical approximation, or choice of complex branches.

Write $[6] = \{1, \dots, 6\}$ and $\mathcal{E} = \binom{[6]}{2}$. Let $\mathcal{P}_2$ be the set of partitions of $[6]$ into three unordered two-element blocks, and let $\mathcal{P}_3$ be the set of partitions into two unordered three-element blocks. Their

cardinalities are

$$|\mathcal{P}_2| = \frac{6!}{(2!)^3 3!} = 15, \quad |\mathcal{P}_3| = \frac{\binom{6}{3}}{2} = 10. \quad (2.2)$$

We call members of $\mathcal{P}_2$ *matchings*. When two matchings have no common block, we call them *block-disjoint*; they still use the same six roots.

For a block $B \subseteq [6]$, define its root polynomial and the complete partition descriptor by

$$p_B(u) = \prod_{i \in B} (u - \alpha_i), \quad D_{\mathcal{B}}(u, v) = \prod_{B \in \mathcal{B}} (v - p_B(u)). \quad (2.3)$$

These are the ordinary polynomial counterparts of the descriptors in `SexticSeparatingInvariants.lean` [7]. The distinction between the variables u and v matters: u enters the root polynomials, whereas v records the unordered collection of blocks.

Lemma 2.1 (A descriptor determines the partition). *For distinct roots, the map $\mathcal{B} \mapsto D_{\mathcal{B}}(u, v)$ is injective on either partition family.*

Proof. First, the map $B \mapsto p_B(u)$ is injective. Indeed, the roots of p_B among $\alpha_1, \dots, \alpha_6$ are precisely the roots indexed by B . Next regard $D_{\mathcal{B}}$ as a monic polynomial in v over the field $K(u)$. Its distinct linear factors are $v - p_B(u)$. Equality of two such products implies equality of their sets of roots in $K(u)$, so it implies equality of the sets of block polynomials. Their injectivity recovers the blocks and hence the partition. This proof also explains why repeated sextic roots must be excluded at this stage. $\square$

The descriptors therefore solve the information-loss problem before specialization. The remaining task is to evaluate them at a pair (u, v) without causing two distinct descriptors to become equal. Arbitrary fixed evaluations need not preserve injectivity. We next expose the unusually small amount of partition-dependent data in the sextic case.

3 Compression to quadratics and linears

3.1 Pair partitions

For a pair $b = \{i, j\}$, write $s_b = \alpha_i + \alpha_j$ and $p_b = \alpha_i \alpha_j$. For a matching P , set

$$\begin{aligned} A_P &= \sum_{b \in P} p_b, \\ B_P &= \sum_{\substack{b, c \in P \\ b \neq c}} s_b p_c, \\ C_P &= \sum_{\{b, c\} \in \binom{P}{2}} p_b p_c, \end{aligned} \quad h_P(t) = A_P t^2 + B_P t - C_P. \quad (3.1)$$

The sum defining B_P is over ordered pairs of different blocks, so it has six summands before expanding the s_b .

Proposition 3.1 (Pair descriptor expansion). *For every matching P ,*

$$\begin{aligned} D_P(u, v) &= v^3 - (3u^2 - e_1 u + A_P) v^2 \\ &\quad + (3u^4 - 2e_1 u^3 + (e_2 + A_P) u^2 - B_P u + C_P) v - f(u). \end{aligned} \quad (3.2)$$

Consequently, if $K_2(t) = 2t^6 + 2e_2t^4 - f(t)$, then

$$D_P(t, 2t^2) = K_2(t) - 2t^2h_P(t). \quad (3.3)$$

The map $P \mapsto h_P(t)$ is injective as a map into $K[t]$.

Proof. Each block polynomial is $u^2 - s_bu + p_b$. Their sum is $3u^2 - e_1u + A_P$. The sum of their pairwise products has leading coefficients 3, $-2e_1$, and its coefficient of u^2 is $2A_P + \sum_{b < c} s_bs_c$. Since the global e_2 is $A_P + \sum_{b < c} s_bs_c$, that coefficient is $e_2 + A_P$. The remaining two coefficients are $-B_P$ and C_P . The product of all three block polynomials is $f(u)$. Expanding the monic cubic in v gives [equation \(3.2\)](#); substituting $u = t, v = 2t^2$ gives [equation \(3.3\)](#).

If two h polynomials agree, their coefficients give equality of A, B, C . The displayed descriptor formula then gives equal descriptors. Apply [lemma 2.1](#). $\square$

3.2 Triple partitions

For a three-element block B , let $e_j(B)$ denote the elementary symmetric polynomial of degree j in its three roots. For $T \in \mathcal{P}_3$, define

$$a_T = \sum_{B \in T} e_2(B), \quad b_T = \sum_{B \in T} e_3(B), \quad k_T(t) = a_Tt - b_T. \quad (3.4)$$

Proposition 3.2 (Triple descriptor expansion). *For every triple partition T ,*

$$D_T(u, v) = v^2 - (2u^3 - e_1u^2 + a_Tu - b_T)v + f(u). \quad (3.5)$$

On $(u, v) = (t, 2t^2)$ this has the form

$$D_T(t, 2t^2) = K_3(t) - 2t^2k_T(t), \quad K_3(t) = 4t^4 - 4t^5 + 2e_1t^4 + f(t). \quad (3.6)$$

The map $T \mapsto k_T(t)$ into $K[t]$ is injective.

Proof. The sum of the two monic cubic block polynomials is $2u^3 - e_1u^2 + a_Tu - b_T$, and their product is $f(u)$. Expanding their quadratic descriptor proves the formula. Equality of k polynomials gives equality of the two descriptor-dependent coefficients, and then [lemma 2.1](#) proves injectivity. $\square$

In particular every difference $h_P - h_Q$ for $P \neq Q$ is a nonzero polynomial of degree at most two; every difference $k_T - k_{T'}$ is nonzero of degree at most one. A first root-counting bound would therefore allow $2\binom{15}{2} + \binom{10}{2} = 255$ bad parameters. The next two sections remove repeated collision factors and reduce this to 195.

Remark 3.3 (The zero parameter and the source coordinate order). The compressed families make sense and may separate at $t = 0$, but the original descriptors on the curve all collapse within each family there, because the multiplier $2t^2$ vanishes. We use positive integer tests when transferring results to the repository. In its Fin2 convention, **X0** is the outer descriptor variable v and **X1** is u . Thus the required parameter is $x(0) = 2t^2$, $x(1) = t$, not the reverse [7].

[write] Checked against the Lean definitions. In `SexticSeparatingInvariants.lean`, `bivariateRootPolynomial` (l. 614) is the product of **X1** minus the roots, and `bivariateDescriptor` (l. 618) the product of **X0** minus those block polynomials, so **X0** = v and **X1** = u as stated; `pairDescriptorValue` (l. 790) evaluates **Xi** at $x(i)$. The Lean search itself ranges over all pairs x , not over the curve; the curve is a subset of its search space. See [section 1.5](#).

4 The two types of pair-partition collision

Two different matchings can share at most one block, since two common pairs force the remaining pair to agree. There are exactly 45 unordered pairs of matchings sharing a block: choose the common pair in 15 ways, then choose two of the three matchings on the four remaining indices. The other $\binom{15}{2} - 45 = 60$ pairs are block-disjoint. Their union is a single alternating six-cycle, since a cycle of length two would be a common block.

For a root pair $e = \{i, j\}$, define

$$q_e(t) = t^2 + s_e t - p_e = t^2 + (\alpha_i + \alpha_j)t - \alpha_i \alpha_j. \quad (4.1)$$

Lemma 4.1 (Common-block collision factor). *If distinct matchings P, Q share the pair e , then*

$$h_P(t) - h_Q(t) = (A_P - A_Q)q_e(t), \quad A_P - A_Q \neq 0. \quad (4.2)$$

Proof. Let the four remaining roots be a, b, c, d . Their two-block pairing affects A through one of

$$z_1 = ab + cd, \quad z_2 = ac + bd, \quad z_3 = ad + bc.$$

The products of the two remaining pair products are always $abcd$. The contribution to B involving only those two blocks is always the third elementary symmetric function of a, b, c, d . Therefore changing the pairing changes B by s_e times the change in A , and changes C by p_e times the change in A . Substitution in [equation \(3.1\)](#) proves the factorization.

The three differences are

$$z_1 - z_2 = (a - d)(b - c), \quad z_1 - z_3 = (a - c)(b - d), \quad z_2 - z_3 = (a - b)(c - d). \quad (4.3)$$

Distinctness of the roots makes every one nonzero. $\square$

Thus the 45 shared-block comparisons do not contribute 90 unrelated exceptional parameters: they contribute roots of only fifteen quadratics. The 60 block-disjoint comparisons each still contribute a quadratic. There are at most $2(15 + 60) = 150$ pair-family failures. Some of these quadratics can have lower degree or share roots after specializing the sextic, so this is a uniform upper bound rather than a claim about every input.

5 Triple collisions as exchanges of two roots

For disjoint root pairs e, e' , set

$$\ell_{e,e'}(t) = (s_e - s_{e'})t - (p_e - p_{e'}). \quad (5.1)$$

Interchanging e, e' changes only the sign. There are $\binom{6}{4} \cdot 3 = 45$ unordered choices of two disjoint pairs.

Lemma 5.1 (Exchange identity). *For the triple partitions*

$$T = \{\{a, b, c\}, \{d, e, f\}\}, \quad T' = \{\{a, b, d\}, \{c, e, f\}\},$$

where letters now denote root values,

$$k_T(t) - k_{T'}(t) = (c - d)((a + b - e - f)t - (ab - ef)). \quad (5.2)$$

Every unordered pair of distinct triple partitions occurs exactly once in this manner, up to the irrelevant orientations, from an unordered pair of disjoint root pairs $\{\{a, b\}, \{c, d\}\}$.

Proof. Subtracting the internal pair sums gives $a_T - a_{T'} = (c - d)(a + b - e - f)$, while subtracting the two triple products gives $b_T - b_{T'} = (c - d)(ab - ef)$. This proves the formula.

For the combinatorial assertion, two distinct partitions can be oriented so that one block from each has two elements in common. Call these a, b ; call the exchanged elements c, d ; the other two elements are e, f . Conversely, two disjoint pairs and their two complementary elements specify exactly the two partitions in the display. Simultaneously switching the chosen blocks or reversing the exchange does not change the unordered data. This establishes the bijection, with 45 elements on either side. $\square$

Lemma 5.2 (No identically vanishing exchange factor). *For disjoint pairs of distinct roots, $\ell_{e,e'}$ is not the zero polynomial.*

Proof. If it vanished identically, the two pair sums and the two pair products would agree. The two monic quadratics $X^2 - s_e X + p_e$ and $X^2 - s_{e'} X + p_{e'}$ would agree, contradicting their disjoint root sets. $\square$

The triple-family failure set is therefore the union of the roots of 45 nonzero linears, and has cardinality at most 45. A nonzero constant factor simply contributes no root.

6 Three invariant obstruction polynomials

Fix total orders on the matchings and on the root pairs, solely to orient differences in products. Define

$$Q_f(t) = \prod_{e \in \mathcal{E}} q_e(t), \quad (6.1)$$

$$U_f(t) = \prod_{\substack{P < Q \\ P \cap Q = \emptyset}} (h_P(t) - h_Q(t)), \quad (6.2)$$

$$L_f(t) = \prod_{\substack{e < e' \\ e \cap e' = \emptyset}} \ell_{e,e'}(t), \quad (6.3)$$

$$E_f(t) = Q_f(t)U_f(t)L_f(t). \quad (6.4)$$

Here $P \cap Q = \emptyset$ refers to their block collections. Reversing a chosen orientation can multiply U_f or L_f by -1 , which changes neither the zero set nor any squared identity below.

It is not automatic that an oriented product of differences is symmetric; the ordinary Vandermonde product is the familiar warning. The following simple sign argument proves what is needed here.

Lemma 6.1 (Graph-product sign test). *Let a finite group act on a finite graph's vertices, permuting quantities w_v . Form an oriented product $\prod_{\{v,w\} \in E} (w_v - w_w)$, with one orientation for each edge. For a group element τ of order two, its sign on this product is $(-1)^r$, where r is the number of edges whose endpoints are interchanged by τ .*

Proof. Edges exchanged with other edges occur in two-element orbits. The two orientation signs in each such orbit multiply to $+1$, since applying τ twice fixes each oriented difference. A fixed edge contributes $+1$ if its endpoints are individually fixed, and -1 if they are interchanged. Multiplying these contributions gives the claim. $\square$

Proposition 6.2 (Symmetry, integrality, and exact failure test). *The expressions Q_f, U_f, L_f are symmetric polynomials with integer coefficients in the six roots and t . They are nonzero for every distinct root tuple and satisfy*

$$\deg Q_f = 30, \quad \deg U_f \leq 120, \quad \deg L_f \leq 45. \quad (6.5)$$

Their coefficients are universal integer polynomials in $e_1, \dots, e_6$. For every $t \in K$,

$$E_f(t) \neq 0 \iff (P \mapsto h_P(t)) \text{ and } (T \mapsto k_T(t)) \text{ are both injective.} \quad (6.6)$$

Proof. The product Q_f is visibly symmetric. For U_f , use the graph with matchings as vertices and block-disjoint pairs as edges. A transposition of two root labels sends any matching either to itself or to a matching sharing its untouched third block. Hence it interchanges the endpoints of no edge in this graph. By lemma 6.1, it fixes U_f . Root transpositions generate S_6 , so U_f is symmetric.

For L_f use the graph on two-element subsets, joined when they are disjoint, with vertex quantity $s_e t - p_e$. A root transposition sends a pair either to itself or to a pair sharing one root. It therefore reverses no disjointness edge, so the same argument proves symmetry of L_f . The fundamental theorem of symmetric polynomials over $\mathbb{Z}[t]$ now yields the integer coefficient expressions in the e_i .

Every factor of Q_f is monic quadratic. Every factor of U_f is nonzero by proposition 3.1, and every factor of L_f is nonzero by lemma 5.2. This proves nonvanishing and the degree bounds. Finally, lemma 4.1 accounts for all shared-block collisions, equation (6.2) accounts for all block-disjoint ones, and lemma 5.1 accounts for all triple collisions. Their scalar root-difference factors are nonzero. This proves equation (6.6). $\square$

Remark 6.3 (Homogeneity and scaling). Assign weight one to every root and to t . Then $h_P, k_T, q_e, \ell_{e,e'}$ have respective total degrees 4, 3, 2, 2. Consequently Q_f, U_f, L_f, E_f have total degrees 30, 240, 90, 360. Scaling all roots and t by c multiplies $E_f(t)$ by c^{360} . These are homogeneous degrees in the roots and parameter together, not the parameter degrees in equation (6.5).

7 Exact discriminant factorizations

Define the compressed resolvents

$$R_{2,f}(Z, t) = \prod_{P \in \mathcal{P}_2} (Z - h_P(t)), \quad R_{3,f}(Z, t) = \prod_{T \in \mathcal{P}_3} (Z - k_T(t)). \quad (7.1)$$

Their coefficients are symmetric integer polynomials in the roots, so they are coefficient-only constructions just as E_f is. Let

$$\Delta_f = \text{disc}(f) = \prod_{1 \leq i < j \leq 6} (\alpha_i - \alpha_j)^2. \quad (7.2)$$

For a monic polynomial with roots $z_1, \dots, z_m$, we use the convention $\text{disc} = \prod_{i < j} (z_i - z_j)^2$.

Theorem 7.1 (Discriminant identities). *As polynomial identities in the roots and t ,*

$$\text{disc}_Z R_{2,f}(Z, t) = \Delta_f^6 Q_f(t)^6 U_f(t)^2, \quad (7.3)$$

$$\text{disc}_Z R_{3,f}(Z, t) = \Delta_f^3 L_f(t)^2. \quad (7.4)$$

In particular the formulas remain valid at nonseparating parameter values.

Proof. The pair-resolvent discriminant is the product of all 105 squared matching differences. The 60 block-disjoint differences give U_f^2 directly. Fix a common pair e . The three unordered comparisons between matchings containing e give, by [lemma 4.1](#),

$$q_e(t)^6 \prod_{1 \leq i < j \leq 3} (z_i - z_j)^2.$$

By [equation \(4.3\)](#), the second factor is exactly the discriminant of the monic quartic whose roots are the other four roots. Multiplying over the fifteen choices of e gives Q_f^6 times the product of these fifteen quartic discriminants. A fixed root difference $(\alpha_i - \alpha_j)^2$ occurs in precisely six of those quartics: choose the excluded pair among the four other roots. Their product is therefore Δ_f^6 , proving [equation \(7.3\)](#).

For triples, [lemma 5.1](#) gives one squared linear factor for each unordered pair of disjoint root pairs, producing L_f^2 . The two leftover roots are the exchanged roots. For a fixed exchanged pair there are three pairings of the other four roots, so its squared difference appears three times. The product of the root-difference factors is Δ_f^3 . This proves [equation \(7.4\)](#). No division was used, so these are polynomial identities, not merely identities on the open set of good parameters. $\square$

Corollary 7.2 (Discriminant square classes). *Over any characteristic-zero coefficient field, the pair-resolvent discriminant is a square. If both f and the triple resolvent are separable, the triple-resolvent discriminant has the same square class as Δ_f .*

Proof. The two formulas read

$$\text{disc}_Z R_{2,f} = (\Delta_f^3 Q_f^3 U_f)^2, \quad \text{disc}_Z R_{3,f} = \Delta_f (\Delta_f L_f)^2.$$

In the second assertion all displayed factors are nonzero. $\square$

There is a useful independent permutation explanation. A transposition of root labels fixes three of the fifteen matchings and interchanges the other twelve in six pairs; it acts evenly. It fixes four triple partitions and interchanges the other six in three pairs; it acts oddly. Thus the fifteen-point action has trivial sign character and the ten-point action has the ordinary six-point sign character. The explicit factorizations identify much more than this parity information: they also isolate every collision factor and its multiplicity.

8 Generic irreducibility of the three components

The degree bound is not merely the result of an inefficient factor count. Generically all its factors are needed. In this section take $x_1, \dots, x_6, t$ to be algebraically independent over $\mathbb{Q}$ and put $S = \mathbb{Q}[x_1, \dots, x_6, t]$. Its symmetric subring is

$$S^{S_6} = \mathbb{Q}[e_1, \dots, e_6, t].$$

The symbols Q, U, L now denote the universal polynomials in this ring, prior to any specialization of the sextic coefficients.

8.1 Irreducibility of the individual root-coordinate factors

Lemma 8.1 (Primitive linear-polynomial test). *Let R be a unique factorization domain. If $A, B \in R$ are coprime and $A \neq 0$, then $Ay + B$ is irreducible in $R[y]$.*

Proof. The polynomial is primitive, and is linear, hence irreducible, over the fraction field of R . Gauss's lemma gives irreducibility over R . $\square$

Lemma 8.2 (The three kinds of factor are irreducible). *In S , each q_e , each $\ell_{e,e'}$ for disjoint pairs, and each $h_P - h_Q$ for block-disjoint matchings is irreducible.*

Proof. For q_e , rename its two roots a, b and write

$$q_e = a(t - b) + t(t + b).$$

The coefficients as a linear polynomial in a are coprime: substituting $t = b$ in the second gives $2b^2 \neq 0$. Apply lemma 8.1. For a linear exchange factor, rename its disjoint pairs $\{a, b\}$ and $\{c, d\}$. Then

$$\ell = a(t - b) + (b - c - d)t + cd.$$

Substituting $t = b$ in its constant coefficient gives $(b - c)(b - d) \neq 0$, so the same test applies. Adjoining unused independent variables preserves these irreducibilities.

For the third type, all block-disjoint matching pairs are related by a root permutation, because their union is an alternating six-cycle. It suffices to consider

$$P = (ab)(cd)(ef), \quad Q = (ac)(be)(df).$$

Write $h_P - h_Q = aA + B$, where $A, B \in \mathbb{Q}[b, c, d, e, f, t]$. This is linear in a because every monomial in equation (3.1) uses each root at most once. It is homogeneous of total degree four; hence A is homogeneous of degree three and B of degree four. At $t = 0$, direct expansion gives

$$\begin{aligned} A_0 &= -bcd + bce - bef + cdf, \\ B_0 &= def(b - c). \end{aligned} \tag{8.1}$$

The polynomial A_0 is divisible by none of $d, e, f, b - c$. Indeed its respective substitutions at zero or at $b = c$ are

$$be(c - f), \quad cd(f - b), \quad bc(e - d), \quad c(e - d)(c - f),$$

all nonzero. Therefore $\gcd(A_0, B_0) = 1$.

Suppose an irreducible nonconstant polynomial H divided both A and B . A factor of a homogeneous polynomial can be chosen homogeneous, so H is homogeneous. If $H|_{t=0}$ is nonzero, it has the same positive total degree and divides both A_0 and B_0 , a contradiction. If it is zero, then t divides H , so irreducibility makes H associated to t ; but $A_0 \neq 0$. Thus A, B are coprime. Apply lemma 8.1 with variable a to finish the proof. $\square$

The homogeneity in the last argument is essential: without it, coprime specializations need not imply coprimality of the original polynomials. Here it prevents a nonconstant common homogeneous factor from specializing to a nonzero constant.

8.2 Descent from root coordinates to coefficients

Lemma 8.3 (One-orbit product lemma). *Let a finite group act on a polynomial ring over a field. Suppose a finite collection of pairwise nonassociate irreducible polynomials forms a single orbit up to units, and its product F is invariant. Then F is irreducible in the invariant subring.*

Proof. A factorization $F = AB$ in the invariant subring is also a factorization in the ambient unique factorization domain. The irreducible factors of A therefore form a subset of the factors of F . Invariance makes this subset stable under the group. Transitivity forces it to be empty or the whole set. In the first case A is a unit, and in the second B is a unit. $\square$

Theorem 8.4 (Generic three-factor theorem). *The universal polynomials Q, U, L are pairwise nonassociate irreducible polynomials in $\mathbb{Q}[e_1, \dots, e_6, t]$, and are irreducible in $\mathbb{Q}(e_1, \dots, e_6)[t]$. Their degrees in t are respectively 30, 120, 45. Their product E is squarefree over the generic coefficient field and has degree 195 and nonzero constant coefficient.*

Proof. The individual factors are irreducible by lemma 8.2. The symmetric group is transitive on root pairs, on unordered pairs of disjoint root pairs, and on unordered pairs of block-disjoint matchings. In each collection the individual factors are pairwise nonassociate. For q_e , this follows from its monic leading term and its two root indices. For $\ell_{e,e'}$, the coefficient of t specifies the two pairs with opposite signs, up to their interchange. For $h_P - h_Q$, the coefficient of t^2 is $A_P - A_Q$: its six distinct root-product monomials, with coefficients $+1$ or -1 , specify the two block-disjoint matchings up to interchange. Thus no further associations occur.

Each product is invariant by proposition 6.2. The one-orbit product lemma proves irreducibility in the symmetric subring. The leading coefficient in t of each individual q is one, of each individual h difference is a nonzero $A_P - A_Q$, and of each individual ℓ is a nonzero difference of pair sums. The ambient ring is a domain, so the three parameter degrees are exactly 30, 120, 45. They are consequently pairwise nonassociate. Their irreducibility and positive parameter degree also imply that they are primitive over $\mathbb{Q}[e_1, \dots, e_6]$. Gauss's lemma proves irreducibility over its fraction field.

In characteristic zero every irreducible polynomial of positive degree is separable. The three are pairwise coprime there, so E is squarefree and has degree 195. Its constant coefficient is nonzero: every $q_e(0)$ and $\ell_{e,e'}(0)$ is a nonzero polynomial, and every block-disjoint h difference at zero is nonzero by the representative calculation equation (8.1) and root permutation. $\square$

This theorem is a statement about the generic coefficient field, not about every numerical sextic. After specialization a factor can become reducible, lose degree, or share roots with another factor. None of these possibilities threatens the uniform separator theorem; they only reduce the number of bad parameters. On the open coefficient locus $\Delta_f \neq 0$, the three polynomials describe the three irreducible families of bad specializations over $\mathbb{Q}$. The constructive factor proofs also work after extending the constant field to any characteristic-zero field.

9 The finite separator theorem and its sharpness

Theorem 9.1 (Uniform simultaneous separation). *For any six distinct roots in a characteristic-zero field, at most 195 parameters cause a collision in either compressed family. Therefore any 196 distinct nonzero test parameters contain a simultaneous separator. The integers $1, \dots, 196$ always contain one, and this same parameter separates both original descriptors on $(u, v) = (t, 2t^2)$.*

Proof. By proposition 6.2, the bad parameters are exactly the roots of the nonzero polynomial E_f , which has degree at most $30 + 120 + 45 = 195$. A nonzero polynomial over a field has at most its degree many distinct roots. Characteristic zero makes the positive integers $1, \dots, 196$ distinct and nonzero. Finally use equations (3.3) and (3.6), whose common multiplier is nonzero at these parameters. $\square$

The same proof gives 151 test values for the pair family alone and 46 for the triple family alone. A common parameter is preferable when a single uniform interface is wanted; separate searches can stop sooner on some inputs. There is no requirement that the least good pair parameter agree with the least good triple parameter.

Theorem 9.2 (Sharpness for arbitrary test sets). *The constant 196 in the assertion about every set of distinct nonzero complex test parameters cannot be lowered. There are separable sextics with exactly 195 distinct nonzero bad parameters. Such a sextic can even be chosen split with integer roots.*

Proof. Over the generic coefficient field, [theorem 8.4](#) gives a degree-195 squarefree polynomial with nonzero constant coefficient. Thus its leading coefficient, constant coefficient, and discriminant in t are all nonzero. Multiply these coefficient polynomials by Δ_f and substitute the elementary symmetric functions of six independent root variables. The result is still a nonzero polynomial in those variables, because the symmetric-polynomial substitution is injective.

The integer lattice is Zariski dense in affine six-space: a nonzero polynomial over $\mathbb{Q}$ cannot vanish on all integer tuples, as follows by induction on the number of variables and the one-variable root bound. Choose an integer tuple where the displayed product is nonzero. Its roots are distinct, and its E_f has degree 195, no zero root, and no repeated root. Over $\mathbb{C}$ it has exactly 195 distinct nonzero roots. These are all bad by [equation \(6.6\)](#); taking them as the test set disproves any 195-point guarantee of the stated kind. $\square$

9.1 An explicit sharpness certificate

The existence proof above is entirely algebraic. The accompanying code also gives a concrete witness with roots

$$1, 4, 10, 23, 51, 109. \quad (9.1)$$

For the orientation used in the verifier, reduction modulo $p = 1,000,003$ gives

$$\deg \bar{E}_f = 195, \quad \bar{E}_f(0) = 521291, \quad \text{lc}(\bar{E}_f) = 670728. \quad (9.2)$$

The file `checks/sharpness_certificate.json`¹ stores the coefficients of $\bar{E}_f$ and polynomials A, B satisfying the exact identity

$$A\bar{E}_f + B\bar{E}'_f = 1 \quad \text{in } \mathbb{F}_p[t]. \quad (9.3)$$

The verifier checks primality of p by trial division and checks this identity by finite polynomial arithmetic. Degree preservation, the nonzero constant coefficient, and [equation \(9.3\)](#) imply that the integer polynomial E_f has degree 195 and 195 distinct nonzero complex roots. This is a reproducible computational certificate, not a Lean-checked one. It is supplementary: [theorem 9.2](#) does not rely on its execution.

9.2 The initial integer interval is a different problem

Define N_{int} to be the least positive integer N such that $1, \dots, N$ contains a simultaneous separator for every distinct complex root tuple. The sharpness proof using complex roots of E_f does *not* prove $N_{\text{int}} = 196$.

Proposition 9.3 (An eight-point lower bound). *For this initial-interval problem,*

$$8 \leq N_{\text{int}} \leq 196. \quad (9.4)$$

¹[write] Shipped as `data/sharpness_certificate.json`; the delivered `checks/` directory is `data/` here.

Proof. The upper bound is [theorem 9.1](#). For the lower bound use the distinct integer roots $(-3, 0, 1, 2, 3, 4)$, whose polynomial is

$$X^6 - 7X^5 + 5X^4 + 55X^3 - 126X^2 + 72X.$$

Direct substitution in [equations \(3.1\)](#) and [\(3.4\)](#) gives the following exact numbers of distinct values:

t	1	2	3	4	5	6	7	8
$ \{h_P(t)\} $	12	13	13	14	15	15	15	15
$ \{k_T(t)\} $	10	8	9	10	9	8	9	10

Thus none of $1, \dots, 7$ works. For additional explicit checks, the sorted pair values at $t = 8$ are

$$\begin{aligned} &-1128, -1044, -870, -792, -724, -486, -432, -424, \\ &72, 90, 192, 396, 776, 816, 936, \end{aligned}$$

and the sorted triple values are

$$-32, -28, -22, -20, -14, 1, 32, 46, 92, 160.$$

They are all distinct within their respective families. □

Proposition 9.4 (No universal single positive parameter). *For each fixed positive integer t_0 , there is a split separable integer sextic for which t_0 fails to separate the pair family.*

Proof. Take roots $0, t_0, 2t_0, 3t_0, 4t_0, 5t_0$. For the common pair $\{2t_0, 3t_0\}$, its quadratic satisfies $q(t_0) = t_0^2 + 5t_0^2 - 6t_0^2 = 0$. Any two different matchings containing that pair collide by [lemma 4.1](#). □

The example is reducible. It does not establish that a fixed parameter must fail on some irreducible rational sextic; that is a separate question.

10 Why the separated resolvents decide an irreducible sextic

The group-theoretic criterion in this section is classical. We include its proof to show exactly what the new separator supplies and what it does not. It is the criterion underlying the degree-fifteen and degree-ten sextic resolvents in the literature and the repository [\[2, 6\]](#). Let $f \in \mathbb{Q}[X]$ now be irreducible of degree six, let M be its splitting field, and view $G = \text{Gal}(M/\mathbb{Q})$ as a transitive subgroup of S_6 .

Lemma 10.1 (Solvable primitive groups have prime-power degree). *A finite solvable group acting faithfully and primitively on a set with more than one element has degree p^r for some prime p .*

Proof. Choose a minimal nontrivial normal subgroup N of G . Because G is solvable, N is elementary abelian. Here is the standard short justification: a last nontrivial term of the derived series of N is an abelian characteristic subgroup of N , hence normal in G , so minimality makes N abelian. Its primary components are characteristic, so only one prime p occurs. The characteristic subgroup of elements killed by p is nontrivial and hence all of N .

The orbits of a normal subgroup form a G -invariant partition. Primitivity and faithfulness force N to be transitive: otherwise its orbits are singletons, making N act trivially. An abelian transitive permutation group is regular. Indeed, an element fixing one point fixes every point by commuting with all elements that carry it to another point. Faithfulness then makes the stabilizer trivial. Thus the degree is $|N| = p^r$. □

Proposition 10.2 (The two block systems are complete). *For a transitive subgroup $G \leq S_6$, the following are equivalent: G is solvable; G is imprimitive; G preserves a partition in $\mathcal{P}_2$ or in $\mathcal{P}_3$.*

Proof. A solvable primitive group cannot have degree six by [lemma 10.1](#). Thus solvability implies imprimitivity. For a transitive action, the blocks in a nontrivial block system have a common size properly dividing six, so that size is two or three. This proves the next implication.

The full stabilizer of a pair partition is $S_2 \wr S_3$, an extension of S_2^3 by S_3 . The full stabilizer of a triple partition is $S_3 \wr S_2$, an extension of S_3^2 by S_2 . Both are solvable, since finite direct products and extensions of solvable groups are solvable. Any subgroup preserving the corresponding partition is a subgroup of its stabilizer and is therefore solvable. $\square$

Theorem 10.3 (Separated rational-root criterion). *Let $f \in \mathbb{Q}[X]$ be an irreducible monic sextic, and let $t \in \mathbb{Q}$ be a simultaneous separator. Then f is solvable by radicals if and only if at least one of $R_{2,f}(Z, t)$ and $R_{3,f}(Z, t)$ has a rational root.*

Proof. For a matching P and $\sigma \in G$, coefficient equivariance gives $\sigma(h_P(t)) = h_{\sigma P}(t)$, since t is rational. If P is fixed by G , then $h_P(t)$ is fixed by G and lies in $\mathbb{Q}$. Conversely, if $h_P(t)$ is rational, then $h_{\sigma P}(t) = h_P(t)$ for every σ , and separation gives $\sigma P = P$. A rational root of the pair resolvent is one of its explicit values, so it is equivalent to a G -fixed pair partition. The same proof applies to k_T and the triple resolvent. Now use [proposition 10.2](#) and the classical equivalence between solvability by radicals and solvability of the Galois group. $\square$

Separation is not a cosmetic condition in the converse. Without it, $h_{\sigma P}(t) = h_P(t)$ would not imply $\sigma P = P$. A specialized value could be fixed while its underlying partition moves among colliding partitions. This is precisely the logical failure prevented by the nonvanishing certificate $E_f(t) \neq 0$.

[write] **Added 29 September 2026, batch 57.** For the Boolean decision on irreducible sextics separation can be bypassed. Part II proves ([theorem 19.1](#)) that f is solvable by radicals if and only if $R_{3,f}(Z, u)$ has a base-field root at one arbitrary u , or $R_{2,f}(Z, t_j)$ has one at each of three distinct t_j . In the nonsolvable case a rational root of $R_{2,f}(Z, t)$ is the common value $(e_2 t^2 + 3e_3 t - e_4)/5$ of the unique invariant pentad of matchings, with multiplicity five, and occurs at no more than two parameters ([theorems 23.5](#) and [24.1](#) and [corollary 23.7](#)). The paragraph above stays true of the proof of [theorem 10.3](#). Part II does not improve the 196-point separation bound, and it writes the resolvents with the parameter first, $R_{2,f}(t, Z)$.

For a monic integer sextic, both resolvents are monic integer polynomials in Z . Any rational root is therefore an integer, so the final condition can be decided using a finite integer interval. No splitting field or numerical root approximation needs to occur in the returned-data algorithm.

11 A bounded coefficient-only decision architecture

11.1 Computing the invariant coefficients without roots

A root-coordinate definition can hide a computational oracle unless the coefficient bridge is made explicit. Here the bridge is the constructive fundamental theorem of symmetric polynomials. For each fixed power of Z in [equation \(7.1\)](#), its coefficient is a fixed symmetric polynomial in six roots with coefficients in $\mathbb{Z}[t]$. It can be reduced to a fixed polynomial in $e_1, \dots, e_6, t$ using integer arithmetic only.

For completeness, the reduction is finite in a particularly elementary sense. Order root monomials lexicographically with $x_1 > \dots > x_6$. The leading exponent vector of a symmetric polynomial has the form

$\lambda_1 \geq \dots \geq \lambda_6$. The product

$$e_1^{\lambda_1 - \lambda_2} e_2^{\lambda_2 - \lambda_3} \dots e_5^{\lambda_5 - \lambda_6} e_6^{\lambda_6} \quad (11.1)$$

has exactly that leading root monomial, with coefficient one. Subtract its multiple by the leading coefficient, which is a polynomial in t . The leading monomial strictly decreases and the total root degree never increases. There are only finitely many root monomials within the original degree bound, so the process terminates with an expression over $\mathbb{Z}[e_1, \dots, e_6, t]$. This proof gives a finite symbolic construction, not a choice of an unspecified algebraic root.

For the coefficient of Z^{15-j} in $R_{2,f}$, the root degree is at most $4j$ and parameter degree at most $2j$. For the coefficient of Z^{10-j} in $R_{3,f}$, they are at most $3j$ and j . Hence degrees at most 60 and 30 in the root variables suffice for the two complete coefficient tables. These are fixed bounds, independent of the input height. The analogous bound for E is 360 in the root variables and 195 in t .

The article does not distribute expanded versions of these potentially large universal tables. Their construction and their evaluation are mathematically specified by the fixed finite products and the reduction above. The supplied executable code is instead a verifier on exact root tuples. This distinction is important when assessing what has and has not been implemented.

11.2 Monicization and bounded factor search

Let the input be an integer polynomial $g(X) = a_6 X^6 + \dots + a_0$, with $a_6 \neq 0$. Define

$$f(Y) = a_6^5 g(Y/a_6) = Y^6 + \sum_{i=0}^5 a_i a_6^{5-i} Y^i. \quad (11.2)$$

This is monic integral, and its roots are a_6 times those of g . Multiplication and division by the nonzero rational a_6 preserve the radical closure of $\mathbb{Q}$, so this transformation preserves all-root radical solvability. The repository already provides this semantic bridge [8].

Write H for the maximum absolute value of the six lower coefficients of f , and let $R = 1 + H$. Every complex root of f has absolute value at most R . For example, if $|z| > 1 + H$, the geometric sum estimate gives

$$H \sum_{i=0}^5 |z|^i < |z|^6,$$

so the leading term cannot be canceled by the others. If a monic degree- d integer polynomial divides f , its roots are a submultiset of these roots, and the coefficient at distance j below its leading term has absolute value at most

$$\binom{d}{j} R^j. \quad (11.3)$$

Gauss's lemma implies that rational reducibility of a monic integer polynomial has a monic integer factor. Any proper factorization of a sextic has a factor of degree at most three. Thus enumeration of the finite coefficient boxes in [equation \(11.3\)](#), for $d = 1, 2, 3$, completely decides reducibility.

A degree-two or degree-three factor leaves a quotient of degree at most four; then every root is radical. A degree-one factor leaves a quintic, whose radical solvability must actually be tested. In particular, one must not accept every reducible sextic: a nonsolvable quintic multiplied by a linear factor is the basic counterexample. The existing primitive-recursive quintic decision can be reused [4, 10]. Repeated factors are handled by these reducible branches; the irreducible branch is automatically separable in characteristic zero.

11.3 The algorithm

The following is a mathematical specification of a finite coefficient algorithm. A default return on malformed input makes it a total Boolean on seven encoded integers; the target predicate includes exact degree six.

Bounded sextic radical criterion.

1. Reject if $a_6 = 0$. Otherwise form the monic integer polynomial f in [equation \(11.2\)](#) and the root bound R .
2. Search the explicit boxes for a monic factor of degree two or three. If one exists, accept.
3. Search the explicit interval for a linear factor. If one exists, choose the first in a fixed order, form its quintic quotient, and return the existing primitive-recursive quintic criterion.
4. The remaining input is irreducible. For $t = 1, \dots, 196$, evaluate the universal coefficient polynomials of the two compressed resolvents. Choose the first t for which both are squarefree, or, equivalently, for which $E_f(t) \neq 0$.
5. Accept exactly when one of the two resulting monic integer resolvents has an integer root. For each resolvent, test all integers in its Cauchy interval $[-(1 + H_R), 1 + H_R]$, where H_R is the maximum absolute value of its lower coefficients.

Squarefreeness in step 4 can be tested by a rational polynomial gcd with the derivative, by a fixed-size resultant determinant, or by the obstruction value itself. For an especially direct primitive-recursive implementation, the fixed-size determinant avoids any concern about a hidden unbounded gcd loop. The resolvent degrees are always fifteen and ten.

Theorem 11.1 (Bounded architecture and correctness). *The coefficient procedure above has a primitive-recursive Boolean realization and accepts exactly the integer inputs of degree six whose complex roots are all expressible by radicals over $\mathbb{Q}$.*

Proof. Monicization is integer arithmetic with fixed exponents. The factor boxes have primitive-recursive bounds, and polynomial division for their fixed degrees is a finite coefficient computation. The correctness of their branches follows from [equation \(11.3\)](#) and the degree arguments above. The quintic subroutine is the established primitive-recursive criterion, not an unbounded search for radical expressions.

On the irreducible branch, [theorem 9.1](#) guarantees that step 4 finds a parameter. Each coefficient is the evaluation of a fixed integer polynomial. Evaluation, fixed-size determinants, a loop of length 196, and bounded selection are primitive recursive under the usual effective encodings of integers and finite tuples. Step 5 is a bounded integer search, and is complete because the resolvents are monic and the Cauchy bound includes all their integer roots. Its correctness is [theorem 10.3](#). Composing these primitive-recursive operations proves the claim. $\square$

This theorem concerns a mathematical realization. It does not assert that the existing Lean function has already been changed, that a `Primrec` proof has been accepted by the kernel, or that the attached verifier is this coefficient solver. A source-level integration must additionally supply the coefficient-evaluation lemmas and bounded-selector code discussed in [section 14](#).

Nor does the naive factor and integer-root enumeration above claim polynomial bit complexity. Its bounds depend polynomially on coefficient *magnitudes*, which can still give exponential running time in their bit lengths. The stronger general polynomial-time theorem is classical [\[1\]](#); our objective is a small, transparent invariant search with a tractable verification boundary.

12 A separator determined by coefficient height alone

The finite search has only 196 candidates, but it is also possible to give one coefficient-dependent parameter that is guaranteed to work without any separation test. It is useful as a termination certificate or a comparison construction, not as a recommendation for practical evaluation.

For a polynomial $P(t) = \sum c_i t^i$ over $\mathbb{C}$, write $\|P\|_1 = \sum |c_i|$. This norm is submultiplicative. Let $f \in \mathbb{Z}[X]$ be monic and separable, put $R = 1 + H$ as above, and use $|\alpha_i| \leq R$. Then

$$\begin{aligned} \|q_e\|_1 &\leq 1 + 2R + R^2 \leq 4R^2, \\ \|h_P - h_Q\|_1 &\leq 6R^2 + 24R^3 + 6R^4 \leq 36R^4, \\ \|\ell_{e,e'}\|_1 &\leq 4R + 2R^2 \leq 6R^2. \end{aligned} \tag{12.1}$$

For the middle estimate, $|A_P| \leq 3R^2$, $|B_P| \leq 12R^3$, and $|C_P| \leq 3R^4$; subtracting two such quantities doubles each bound. Multiplying the bounds for fifteen, sixty, and forty-five factors gives

$$\|E_f\|_1 \leq 4^{15} 6^{165} R^{360} =: M(H). \tag{12.2}$$

Theorem 12.1 (A search-free height separator). *The positive integer*

$$t_* = 4^{15} 6^{165} (1 + H)^{360} + 2 \tag{12.3}$$

separates both compressed families and both original descriptors on the curve.

Proof. The nonzero polynomial E_f has integer coefficients, so its leading coefficient has absolute value at least one. Its Cauchy root bound is at most $1 + M(H)$ by [equation \(12.2\)](#). Hence $E_f(t_*) \neq 0$. Apply [equation \(6.6\)](#) and the nonzero-multiplier identities. $\square$

Although t_* is numerically enormous, its bit length is $O(1 + \log(1 + H))$ with a large absolute constant. Its role is to show that even separator selection itself can be replaced by an explicit arithmetic expression. The first successful candidate in $1, \dots, 196$ will generally be much smaller and avoid unnecessary coefficient growth.

13 Exact checks, certificates, and their limits

The attached computations are designed to test local algebra, combinatorial counts, discriminant multiplicities, symmetry signs, and explicit examples. They are not used as evidence that an unproved general statement should be accepted. Each general theorem above has a written argument, including the irreducibility and arbitrary-test-set sharpness results.

13.1 Local symbolic identities

The script code/verify.py uses SymPy to expand and check four universal identities: the pair and triple descriptor expansions, the common-block factorization, and the triple exchange factorization. These checks involve independent root symbols, rather than approximate roots of sample polynomials. They therefore test the precise signs and coefficients in [equations \(3.2\), \(3.5\), \(4.2\) and \(5.2\)](#).

13.2 Integer arithmetic checks

The same script enumerates the fifteen matchings and ten triple partitions without a stored table. It checks the $45 + 60$ decomposition of matching comparisons and the 45 disjoint-root-pair comparisons. For each root transposition it checks that no edge of the block-disjoint matching graph is reversed, matching the sign argument in [lemma 6.1](#).

For four distinct integer root tuples and every $t = 0, \dots, 16$, the script checks both discriminant identities using exact integer products, and verifies the compressed-to-descriptor identities and the exact obstruction test. This yields 68 parameter cases, each testing both resolvent discriminants. It also checks invariance of the three factors under all fifteen root transpositions for those four tuples, giving 60 exact symmetry cases.

It exhausts all $\binom{11}{6} = 462$ six-element subsets of $\{-3, -2, \dots, 7\}$ and computes their first positive simultaneous separator. The resulting distribution is

First separator	1	2	3	4	5	6	7	8
Number of tuples	93	83	89	125	48	15	8	1

These finite data do not support a claim that eight works universally. They provide reproducible examples, including the lower-bound witness in [proposition 9.3](#). Twenty scaled examples additionally check the failure of fixed parameters described in [proposition 9.4](#).

13.3 A modular certificate for the sharp example

The independent standard-library script `code/sharpness_certificate.py` constructs E_f modulo 1,000,003 for [equation \(9.1\)](#). It uses only its 120 explicit factors, computes an extended Euclidean identity with its derivative, and checks [equation \(9.3\)](#). The certificate stores every coefficient in ascending order, so an independent implementation can verify the identity by multiplication and addition without repeating the Euclidean algorithm.

Because reduction preserves the degree and gives gcd one, the characteristic-zero polynomial is squarefree. Equivalently, the resultant with its derivative is nonzero modulo the prime, hence is a nonzero integer. This proves that the example reaches 195 distinct nonzero complex failures for this parameterization. It does not imply that any of those failures, let alone the first 195 positive integers, are all integers or all real.

13.4 Verification status

Both scripts were executed successfully for this report. The package records the exact results, the complete modular identity, and the scripts needed to reproduce them. There was no Lean or Rocq execution for the new results, no rebuild of the repository, and no benchmark against production Galois software. The code supplied is an exact verifier and root-level prototype, not a completed coefficient-only radical solver.

14 A precise route back into ProveIt

All repository statements here refer to snapshot `e2b1f016a94102663f12b970e6dd434229f90d01`. The relevant source directory is `Algebra/PolynomialFormulas/Lean/PolynomialFormulas/`. A subsequent repository version may have changed its interfaces or already closed some of these obligations.

14.1 Reusable existing boundaries

The source already provides descriptor injectivity and evaluated-resolvent separability equivalences in `SexticSeparatingInvariants.lean`; the irreducible decision combines the chosen parameters with the Galois criterion in `SexticIrreducibleDecision.lean`. The final all-roots predicate and both computable and primitive-recursive criterion interfaces are in `SexticRadicalComputability.lean`. Monicization is separated cleanly into `SexticRadicalSemantics.lean` [7, 6, 9, 8].

The reducible branch also contains a smaller search issue: `linearTotalCode` uses `Nat.find`, even though a finite symmetric integer-code bound is already present in `linearWitnessB`. A complete primitive-recursive integration must replace or bound that selector as well, not stop after fixing the pair and triple search [10].

[write] Two files this section does not name. The unguarded pair and triple searches, `pairSeparatingCode` and `tripleSeparatingCode`, are in `SexticSeparatingSearch.lean` [11]; `SexticIrreducibleDecision.lean` wraps them in the total searches `pairTotalCode` and `tripleTotalCode`. A third unbounded search, `elementaryCode` in `SexticSparseSymmetricSearch.lean` [12], finds the elementary-symmetric certificates for the resolvent coefficients and the collision products. Declarations and lines are in [section 1.5](#).

14.2 Suggested module sequence

A useful first module, tentatively `SexticCompressedDescriptors.lean`, would define A, B, C, a, b, h, k and prove [equations \(3.2\)](#) and [\(3.5\)](#). These are finite ring identities, followed by short applications of the existing descriptor injectivity theorem. Its main public result should identify the current parameter $x(0) = 2t^2, x(1) = t$ with the compressed values through the common nonzero affine transformation.

A second module, `SexticCollisionFactors.lean`, would formalize the shared-block/exchange identities and the 45/60/45 finite partition counts. It can obtain the 195 bad-value bound directly from a product over the finite factors, without first formalizing generic irreducibility or the full discriminant formulas. Thus the shortest route to the bounded selector is substantially smaller than the whole article.

A third module, `SexticBoundedSeparatingSearch.lean`, would prove that one of $1, \dots, 196$ works and implement a fixed-length list scan. Its return specification should establish simultaneous injectivity, followed by the existing resolvent correctness bridge. A total fallback value can be present in the definition, but its unreachable status on separable inputs must be proved rather than assumed.

The computational module should then prove primitive recursiveness of each fixed universal coefficient evaluation, of the squarefreeness test, and of the bounded selector. One legitimate shortcut is to specialize at each of the 196 constant parameters and unroll the finite search, provided the coefficient evaluators themselves have primitive-recursive proofs. Merely replacing `Nat.find` by a bounded loop around a function known only to be computable would not establish `Primrec`.

Finally, replace the linear-factor selector by its existing bounded interval, reuse the primitive-recursive quintic branch, and compose the final criterion. The independent Rocq route can use the same mathematical bound and compact collision products, but no claim of a completed port is made here.

14.3 Separate proof products from optional refinements

The exact discriminant identities, generic irreducibility, and the 195-failure sharpness example deserve their own modules. They make the mathematical story stronger, but they are not prerequisites for a correct bounded sextic decision. Keeping them separate would preserve a small trusted dependency chain for the computational endpoint.

A sensible audit should report four distinct endpoints: the universal mathematical separator theorem; the correctness of the bounded returned parameter; a primitive-recursive coefficient function; and the final all-roots radical predicate. Exact sample evaluations are a fifth product, not a substitute for any of the first four.

15 A reusable general finite-separation principle

The sextic construction is special in its low degrees and factorization, but the idea of converting polynomial descriptors to a finite hitting set has an elementary general form. The following statement explains why an existence proof by avoiding finitely many polynomial equalities often admits an explicit bounded counterpart.

Proposition 15.1 (Kronecker-curve separation). *Let $F_1(u, v), \dots, F_m(u, v)$ be distinct polynomials over a characteristic-zero field, each having u -degree at most a and v -degree at most b . Put*

$$G_i(t) = F_i(t, t^{a+1}), \quad D = a + b(a + 1).$$

Then the G_i are distinct, have degree at most D , and among any $D \binom{m}{2} + 1$ distinct parameter values there is one at which all their values are different.

Proof. A monomial $u^i v^j$ becomes $t^{i+(a+1)j}$. For $0 \leq i \leq a$, the exponent uniquely determines i, j by division with remainder by $a + 1$. Thus the substitution is injective on the stated polynomial space and preserves nonzero differences. The product of the $\binom{m}{2}$ differences is nonzero of degree at most $D \binom{m}{2}$, and the root bound applies. $\square$

This is a standard monomial-encoding argument, not a novelty claim. Its contrast with the sextic result is instructive. A general curve must preserve all descriptor coefficients. Here we first subtract the partition-independent part, use the coefficient relations to compress the family, and then recognize common collision factors. The result is a quadratic curve, a low-degree obstruction, and an exact generic factor count rather than a large undifferentiated degree estimate.

For other resolvent actions the promising order of work is therefore: identify a complete descriptor; isolate its varying coefficients; find a low-degree curve faithful to those coefficients; organize collision pairs by group orbits; and only then use a root bound. The graph-product sign test and the one-orbit product lemma can be reused independently of the sextic setting.

16 Further research questions and priorities

The results leave several concrete problems. They fall into three groups: sharp arithmetic separation, the geometry of the obstruction, and certified implementation. The distinction between the solved arbitrary-set bound and the unresolved integer-interval bound should be retained in all follow-up work.

Question 16.1 (The optimal initial integer interval). Determine N_{int} , currently bracketed by $8 \leq N_{\text{int}} \leq 196$ in this report. A proof that eight suffices would require new control over simultaneous integer roots of the three factors, not just an improvement in generic complex root counting. Conversely, construct a distinct root tuple for which 1, $\dots$, 8 all fail.

A practical first experiment is an exact search over larger integer and rational root sets, recording which factor accounts for each failed parameter. The combinatorial pattern of overlaps may be more informative than the maximum alone. Any proposed universal improvement should be proved independently of a finite sample.

Question 16.2 (Short fixed test sets not forming an interval). What is the smallest cardinality of a fixed set $S \subset \mathbb{Z}_{>0}$ that separates every distinct complex root tuple? The sharpness theorem for arbitrary input-dependent complex test sets does not answer this question. Can a sparse integer set outperform an initial interval substantially?

Question 16.3 (Irreducible rational inputs). Do the best uniform bounds decrease when f is required to be irreducible over $\mathbb{Q}$? For a prescribed positive integer t_0 , can one construct an irreducible rational sextic with $E_f(t_0) = 0$? The split counterexamples in this article deliberately do not settle this.

A route through rational points on the three collision hypersurfaces would have to preserve irreducibility of the sextic, rather than merely produce a generic point in root coordinates.

[write] Answered for the decision in Part II, 29 September 2026 (batch 57). For deciding radical solvability, as opposed to separating, irreducible inputs need no uniform range at all: the four fixed tests $u = 1$ and $t = 1, 2, 3$ suffice ([theorem 19.1](#)), over any characteristic-zero base field. Whether the uniform separation bounds decrease under irreducibility, and whether $E_f(t_0) = 0$ is possible for an irreducible rational sextic, remain open. Part II's [theorem 22.2](#) and [lemma 23.2](#) give a partial constraint, noted at the end of [section 18.4](#): for a nonsolvable irreducible sextic, $E_f(t_0) = 0$ at a base-field t_0 forces $U_f(t_0) = 0$.

Question 16.4 (Alternative low-degree curves). Can a different scalar compression or a different curve in descriptor space lower the maximum number of exceptional parameters below 195? The proof of sharpness applies to the particular h, k parameterization, not to every possible separating invariant.

The common-block factorization suggests looking for a curve that collapses more repeated comparisons without making an entire family identical. Any such proposal must still distinguish all partitions for every distinct root tuple, including special symmetric configurations.

Question 16.5 (Geometry of the three components). Describe the intersections, singularities, and branching of the three irreducible hypersurfaces $Q = 0$, $U = 0$, and $L = 0$ over the separable-sextic coefficient locus. Which specializations cause degree loss, repeated parameter roots, or intersections between the components?

The generic irreducibility theorem reduces this from an unspecified collision locus to three explicit orbit products. Their normalizations and projection monodromy are natural next invariants.

Question 16.6 (Galois groups of the obstruction factors). Determine the Galois groups over $\mathbb{Q}(e_1, \dots, e_6)$ of the degree-30, degree-120, and degree-45 parameter polynomials. Their factorizations into quadratics and linears over the generic sextic splitting field provide strong constraints, but do not by themselves determine the full groups.

Question 16.7 (Compact coefficient circuits). Construct certified straight-line programs for the two resolvents or for $E_f(t)$ that avoid expanding large symmetric-polynomial tables. Compare Newton identities, traces in finite algebras, and modular reconstruction, with exact operation counts and coefficient-height estimates.

A useful first target is not the entire final solver but an independently checkable evaluator for the obstruction at a supplied integer parameter.

[write] Partly addressed in Part II, 29 September 2026 (batch 57). Part II gives coefficient-only formulas for both resolvents from the 45- and 20-dimensional compound matrices of the companion matrix ([theorems 26.1](#) and [26.2](#)), with exact power extraction, and an implementation. It gives no operation counts or height estimates and no evaluator for E_f ; its Research question [30.6](#) asks for smaller realizations.

Question 16.8 (End-to-end formal primitive recursiveness). Implement the module sequence in [section 14](#), close the bounded linear-selector issue, and obtain a kernel-checked **Primrec** theorem for the complete coefficient criterion. Port the same bounded search to the independent Rocq development.

The decisive artifact would be a verified executable Boolean and its all-complex-roots specification, not only a classical existence theorem for a suitable natural number.

[write] **Added 29 September 2026, batch 57.** On the irreducible branch Part II’s [proposition 27.1](#) gives a primitive-recursive realization with no separating search at all, as a written argument. The kernel-checked Primrec theorem asked for here remains open, and nothing in Part II is formalized ([section 18.3](#)).

Question 16.9 (Modular certificates and practical performance). Can a small collection of finite-field evaluations certify good parameters and rational-root decisions substantially faster than direct integer expansion? Establish explicit reconstruction bounds and fallback conditions so that modular success is a certificate rather than a probabilistic guess.

Question 16.10 (Positive characteristic and field size). Determine the precise characteristics and finite-field sizes for which analogues hold. The compressed formulas and many collision identities survive algebraically, but the multiplier $2t^2$, the distinctness of integer parameters, and generic irreducibility arguments must be revisited. Extension-field parameters may be necessary.

Question 16.11 (Higher-degree block systems). For partitions into equal-sized blocks in higher degrees, classify collision orbits and search for compressed descriptors with similarly small invariant products. Which actions admit a graph-product factorization and irreducible orbit factors of manageable degree?

Question 16.12 (From a decision to explicit radical expressions). Use a certified rational resolvent root to recover a fixed block system and then construct branch-compatible radical expressions. A Boolean solvability certificate is not yet an explicit all-roots formula; the recovery and radical-tower interfaces require separate proofs.

17 Conclusion

The repository’s unbounded separating search can be replaced by a mathematically explicit finite mechanism. The central reason is not a general computability theorem, but a small amount of partition-dependent information: three coefficients for pair partitions, two for triple partitions, and a quadratic curve that preserves them. The resulting collision factors have enough structure to yield an exact discriminant factorization and a generic irreducibility theorem.

This resolves the bounded-search obligation at the written-proof level and identifies a precise next formalization target. It also settles a sharpness question for arbitrary complex test sets while leaving the more arithmetic positive-integer problem genuinely open. Keeping these scopes separate is necessary both for mathematical accuracy and for a useful research program.

[write] **Status of the obligation in ProveIt.** In the Lean and Rocq developments the obligation stays open: the separating parameters are still found by unbounded minimization, and formalizing [theorem 9.1](#) would bound only the parameter selector ([section 1.5](#)).

A Reproducibility and package contents

The archive contains the self-contained source `article.tex`, its compiled `article.pdf`, `README.md`, a build script, two Python scripts, the exact verification report, and the modular Bézout certificate. Bibliographic entries are embedded in the TeX source; no separate bibliography tool is required. A checksum file records the packaged artifacts.

Compile the document by running

`./build.sh`

from the package directory. The script uses `pdflatex` three times and writes the PDF in the same directory. The source uses commonly available TeX Live packages, including `newtx`, `microtype`, and `tcolorbox`.

Run the checks with

```
python code/verify.py
python code/sharpness_certificate.py
```

The first script needs SymPy for its four local symbolic identities. The second uses only the Python standard library and the combinatorial and integer-arithmetic functions defined in `verify.py`; importing that module does not import SymPy or run its test suite. The certificate polynomials use ascending coefficient order. The default scripts use assertions for test failures and should be run without Python’s optimization flag.

[write] The shipped layout. This appendix describes the delivered archive. In the collection the report directory holds `article.tex`, a build of it as `article.pdf`, a report `README.md` (not the delivered one), `code/verify.py`, `code/sharpness_certificate.py`, `code/build.sh`, and `data/verification.json`, `data/sharpness_certificate.json`, `data/environment.json`, `data/requirements.txt`; the delivered PDF, README and checksum file are not shipped. `code/build.sh` changes to its own directory, where there is no `article.tex`; build instead with `latexmk-pdf` from a copy of the report directory. Each script writes its output into a directory `checks/` next to the `code/` directory that contains it, so run them on a copy of `code/`, never in place, and compare the new `checks/X` with `data/X`; the README gives the commands.

B Source and claim ledger

The following ledger separates historical dependencies, inspected source status, written results, and executed checks.

Claim or artifact	Basis and status
General radical-solvability decidability	Classical; Landau–Miller proves polynomial-time decidability. Not a new result of this article.
Sextic pair/triple group criterion	Classical; proved again in section 10 to expose the exact role of separation.
Complete block descriptors	Inherited from the inspected repository; injectivity reproved in lemma 2.1 .
Quadratic/linear compression	Written proof in section 3 ; local identities checked symbolically.
Three invariant factors and discriminants	Written proofs in sections 6 and 7 ; exact integer tests supplied.
Generic irreducibility and 196 sharpness	Written proofs in sections 8 and 9 ; explicit sharp example additionally has a modular certificate.
Initial interval optimum	Not determined; this report proves only $8 \leq N_{\text{int}} \leq 196$.
Primitive-recursive coefficient criterion	Mathematical construction and proof in section 11 ; not an executed Lean development or a packaged extracted solver.
Executable checks	Both attached scripts executed successfully; exact outputs in <code>checks/</code> (shipped as <code>data/</code> ; [write]). No floating-point evidence is used.
Historical novelty	Specific contributions are relative to the inspected repository and consulted sources; no exhaustive priority certification.

Part II. Radical-solvability tests without separating sextic resolvents

Sections 1–17 and Appendices A–B form the original report (Part I), unchanged except for dated [write] notes pointing to this addition. Part II was added on 29 September 2026 in batch 57 of ProveIt’s incoming-report intake, from one manuscript written against Part I as printed above. For the Boolean decision on irreducible sextics it bypasses Part I’s separation requirement: one triple test at an arbitrary parameter, or matching tests at three distinct parameters, decide radical solvability (theorem 19.1). This answers Part I’s Research question 16.3 for the decision only, not for the separation bounds. Section 18 records the provenance, fixes the notation and states the formal status of the addition; Sections 19–31 and Appendices C–E are the manuscript.

GALOIS THEORY / FINITE GEOMETRY / EXACT COMPUTATION

Three Plus One Radical-Solvability Tests Without Separating Sextic Resolvents

Pentad rigidity, a two-point exceptional set,
and exact coefficient-only evaluators

Prepared for Vladimir Reshetnikov

September 29, 2026

Abstract (manuscript 03). For an irreducible sextic over a characteristic-zero field, we prove a fixed four-resolvent criterion that does not require separating specialized invariants. One triple-partition test and three matching tests suffice:

radical solvability $\iff$ one triple test succeeds $\vee$ all three matching tests succeed.

The matching parameters may be any three distinct base-field elements; the triple parameter is arbitrary. The proof identifies the entire possible false-positive mechanism. For a nonsolvable sextic, every base-field root of the matching resolvent has multiplicity exactly five, arises from the unique possible invariant one-factorization of K_6 , and equals $(e_2t^2 + 3e_3t - e_4)/5$. Such false positives occur at no more than two parameters. The triple resolvent is separable and has no base-field root at every base-field parameter in the nonsolvable case. We prove these statements without classifying the transitive subgroups of S_6 .

We also give coefficient-only matrix formulas of sizes 45 and 20, with exact perfect-power extraction, and supply a working implementation. Exact checks cover symbolic identities, split-root comparisons, and forty-eight irreducible examples representing all sixteen transitive sextic group types. The final sections discuss bounded arithmetic, repository integration, and questions left open by the exceptional-set theorem.

Status. A research draft with complete written proofs and executable exact checks; unrefereed and not formalized in Lean or Rocq. The proposed contributions are relative to the inspected repository and consulted sources; historical priority is not certified. General decidability and polynomial-time decidability of radical solvability are classical. No claim is made that three matching tests are optimal,

or that a false-positive parameter exists.

18 [write] Provenance, conventions and status of Part II

18.1 One manuscript, one addition

Part II is batch-57 manuscript 03, *Three Plus One: Radical-Solvability Tests Without Separating Sextic Resolvents. Pentad rigidity, a two-point exceptional set, and exact coefficient-only evaluators*, an AI-assisted research article prepared for Vladimir Reshetnikov and dated 29 September 2026 (22 pages, US Letter, as delivered). It arrived as the archive `ProveIt_Three_Plus_One_Resolvents.zip` (11 files; inner directory of the same name, main file `article.tex`) in ProveIt commit `6ad01e9c8` and was placed in `70b39943a`. It was written against ProveIt commit `e47ed8547bd021f9bd85bdcf767c94a2fcb41001` (the arrival commit of batch 56), at which Part I’s `article.tex` had the Git blob `3f52dc51`. That blob was unchanged at the placement commit, so every statement the manuscript makes about “the predecessor report” or “the separator report” refers to Part I as printed above, before the dated notes of this write. The five Lean files it cites (`SexticRadicalDecision.lean`, `SexticSeparatingSearch.lean`, `SexticSparseSymmetricSearch.lean`, `SexticSeparatingInvariants.lean`, `SexticReducibleDecision.lean`) have the same blobs at its pin, at Part I’s pin `e2b1f016a` and at the placement commit. The Git blob it records for its Sympy test-seed file is external to ProveIt and was not checked.

The manuscript’s two programs, build script and four data files are shipped beside Part I’s files with the prefix `02-three-plus-one-` (Part I’s unprefix files count as 01): `code/02-three-plus-one-sextic_resolvents.py`, `code/02-three-plus-one-verify.py`, `code/02-three-plus-one-build.sh`, `data/02-three-plus-one-requirements.txt`, `data/02-three-plus-one-sources.json`, `data/02-three-plus-one-verification.json` and `data/02-three-plus-one-verification.log`, byte-identical to the delivery. Its source, README and PDF are not shipped and survive in the arrival commit; its checksum file `SHA256SUMS` (10 of 10 verified at placement) is not shipped. The manuscript did not see any other batch-57 delivery.

The manuscript contributes, in the order printed here: the question and the three-plus-one criterion (section 19); a self-contained account of the descriptors and compressed resolvents (section 20); the block criterion and the invariant-regular-graph lemma on six vertices (section 21); distinct shifted pair products and the safe triple test (section 22); the shared-edge factorization, the six pentads and the classification of rational matching fibers, with the forced value μ_f (section 23); the exceptional polynomial J_f and the two-point theorem (section 24); the proof of the criterion, early-exit certificates and an orbit-budget bound (section 25); the 45- and 20-dimensional compound-matrix identities and exact power extraction (section 26); the delivered decision function, root boxes and a primitive-recursive realization of the irreducible branch (section 27); its exact checks, including all sixteen transitive group types (section 28); a source map and formalization order (section 29); ten research questions (section 30); its conclusion (Section 31); and three appendices: the matching and pentad tables, the delivered run instructions, and a proof-dependency checklist (Appendices C–E). Every result, proof, example, remark, table, limitation and question of the manuscript is printed.

Where the merge had to choose:

- **Numbering.** The manuscript’s Section n is Section $n + 18$ here, and its Appendices A–C are Appendices C–E. Theorem-like and equation numbers follow the sections: the manuscript’s Theorem 1.1 is Theorem 19.1 and its equation (1.1) is (19.1). Its one table, Table 1 there, is Table 4 here, because the report’s longtables

share the table counter.

- **Front matter.** The manuscript’s title page, abstract and status box open Part II; its table of contents is replaced by the report’s.
- **Overlap with Part I, kept.** The manuscript reproves the algebraic facts it inherits “so the new argument does not assume the correctness of every claim in an unrefereed predecessor”. [Lemma 20.1](#) is [lemma 2.1](#); [proposition 20.2](#) and [\(20.6\)](#) are [propositions 3.1](#) and [3.2](#) with [\(3.3\)](#) and [\(3.6\)](#); [lemma 23.1](#) is [lemma 4.1](#), by the same argument; and [lemma 21.1](#) is [lemma 10.1](#) together with the solvable-iff-block-system part of [proposition 10.2](#), by the same minimal-normal-subgroup argument. They are printed in both parts, with notes here and there. [Theorem 19.1](#) extends Part I’s [theorem 10.3](#): it needs no separating parameter, at the price of four tests instead of one and an irreducibility hypothesis over an arbitrary characteristic-zero field.
- **Bibliography.** Entries both sources cite are printed once: Landau–Miller and Boswell–Glasser (with Part II’s annotation added) and the four Lean files. The manuscript’s key repo–search is `SexticSepa ratingSearch.lean`, whereas Part I’s key repo–search is `SexticIrreducibleDecision.lean`; Part II’s citations of that file therefore point to the entry Part I’s write added for it [11], and its repo–sparse to [12]. The manuscript attributes the Lean files to “ProveIt contributors”, Part I to Vladimir Reshetnikov; Part I’s wording is kept. Its other four entries keep their text under keys prefixed `tpo::`; its entry for the separator report is Part I at the pin.
- **Macros.** The manuscript’s `\M`, `\T` and `\pin` are `\tpoM`, `\tpoT` and `\tpopin` here, and its `\E` is Part I’s `\Edges`; the printed symbols are unchanged. Its four unused operator declarations were not carried over. The `listings` package was added for its two code listings.
- **Shipped names.** The manuscript names its files by their delivered paths. Notes in [section 28](#) and [section D](#) give the shipped names; in particular its `data/verification.json` is `data/02-three-plus-one-verification.json`, and the file `data/verification.json` of this directory is Part I’s.
- **Part I** is unchanged except for the title-page provenance line, a paragraph in [section 1.4](#), a table-of-contents line for Part I, labels on Research questions [16.3](#), [16.7](#) and [16.8](#), dated notes after [theorem 10.3](#) and after those three questions, and the bibliography annotations just described.

18.2 Conventions and notation

No symbol printed in the manuscript was renamed and no normalization was changed. Part II keeps its own notation, which differs from Part I’s at the places listed below; where a false reading is tempting it is printed beside the true one.

Symbol	Meaning in Part II, and readings to avoid
$R_{2,f}(t, Z), R_{3,f}(t, Z)$	The compressed resolvents (20.3) , written with the parameter <i>first</i> ; Part I writes the same polynomials as $R_{2,f}(Z, t), R_{3,f}(Z, t)$ (7.1) . <i>False reading</i> : that Z is the parameter in Part II. In Table 4 , “ $R_3(1)$ ” and “ $R_2(t)$ ” abbreviate $R_{3,f}(1, Z)$ and $R_{2,f}(t, Z)$.
K, L, G	K is the base field over which f is irreducible; it does not contain the roots. In Part I Sections 2–9 the field K contains the roots. L is the splitting field and $G = \text{Gal}(L/K)$; Part I calls the splitting field M (section 10) . <i>False reading</i> : L as Part I’s triple collision factor L_f or the universal L .
u, t	In theorem 19.1 u is the parameter of the triple test and t_1, t_2, t_3 those of the matching tests. In the descriptors $D_{\mathcal{B}}(u, v)$ and on the curve $(u, v) = (t, 2t^2)$ of (20.6) , u is the root variable, as in Part I. The triple test evaluates $k_T(u)$ directly, not on the curve.
$\mathcal{M}, \mathcal{T}, \mathcal{E}$	The matchings, the triple partitions and the fifteen root pairs (edges of K_6): Part I’s $\mathcal{P}_2, \mathcal{P}_3, \mathcal{E}$. <i>False reading</i> : $\mathcal{E}$ as Part I’s obstruction polynomial E_f or the universal E .
edge-disjoint	Matchings with no common edge. Since the blocks of a matching are edges, this is Part I’s “block-disjoint”.

Symbol	Meaning in Part II, and readings to avoid
pentad, Σ	Five pairwise edge-disjoint matchings, a one-factorization of K_6 ; there are six (lemma 23.4). Part I has no pentads.
$\mu_f, J_f, \mathcal{B}_f$	New in Part II: the forced rational value (23.4), the exceptional polynomial (24.1), and the set of false-positive parameters. $\mathcal{B}_f$ is a set of parameters, not a partition $\mathcal{B}$.
r	The fiber size in theorem 23.5, and the degree of an invariant regular graph in lemma 21.2; the number of orbits in proposition 25.1. The matrices $r_1, \dots, r_4$ of (26.2) are unrelated.
$C, S, P, Q, H,$ U, V, D, F	In section 26: the companion matrix, its additive and multiplicative compounds, the matrices of (26.4) and (26.5), $D = (uI_6 - C)^{\wedge 3}$ and $F = f(u)$. They are not C_P , a matching P , Part I's Q_f or universal Q , the height H , U_f , the descriptor variables u, v (the manuscript says so), or descriptors $D_{\mathcal{G}}$. In the power extraction, $A(Z)$, $Q(Z)$ and q_k are polynomials and their coefficients, not A_P , the matrix Q , or q_e .
$H, R = 1 + H$	In section 27, as in Part I's section 11: H is the largest absolute value of the nonleading coefficients of a monic integral sextic and $R = 1 + H$ a root bound.
$q_e, h_P, k_T,$ $A_P, B_P, C_P, a_T, b_T,$ K_2, K_3	As in Part I (equations (3.1), (3.3), (3.4), (3.6) and (4.1)); the formulas agree term by term.

18.3 Formal status in ProveIt

None of Part II is formalized. No `tpo:` label has a Lean or Rocq counterpart. The pentad lemmas, the triple rigidity, the fiber classification, the exceptional-set theorem, theorem 19.1, the matrix identities and proposition 27.1 have written proofs only; proposition 27.1 is a mathematical argument, not a `Primrec` proof. Placement beside a Lean and Rocq development confers no formal status. What the project does prove, in the files the manuscript cites, is listed in section 1.5; the blobs are the same.

What formalizing Part II would give (a formalization target, not a claim). On the irreducible branch the decision needs no separating parameter at all. A Lean proof of theorems 19.1, 22.2, 23.5 and 24.1 against the existing descriptor definitions would let that branch test the four fixed parameters $u = 1$, $t = 1, 2, 3$ (in Lean's order the points $x = (2t^2, t)$ for $t = 1, 2, 3$, the first serving the triple test as well; by (20.6) the descriptor values there are affine images of the compressed ones) instead of calling `pairSeparatingCode` and `tripleSeparatingCode` (`SexticSeparatingSearch.lean`, l. 604, 608), or the guarded `pairTotalCode`, `tripleTotalCode` of `SexticIrreducibleDecision.lean` that wrap them, which the manuscript does not cite. The certificate search `elementaryCode` (`SexticSparseSymmetricSearch.lean`, l. 634) would remain unless the matrix identities theorems 26.1 and 26.2 were also formalized as the coefficient evaluator, and `linearTotalCode` (`SexticReducibleDecision.lean`, l. 72) remains on the reducible branch in any case (section 29 says the same).

18.4 What is not claimed

The manuscript states each of the following, and each is kept where it states it. It is unrefereed and not formalized in Lean or Rocq; the repository was inspected at the pin and not modified or rebuilt; its proofs are conventional written proofs, not imported Lean theorems. Its contributions are proposed relative to the inspected repository and the consulted sources; historical priority is not certified, and novelty is not inferred from the absence of a statement in the repository. General decidability, and polynomial-time decidability, of radical solvability are classical (Landau and Miller), as is the sextic block criterion (Boswell and Glasser);

[proposition 27.1](#) is not a new decidability result, claims no polynomial bit complexity for the code and no speed advantage over production Galois algorithms, and its root boxes are a coarse termination bound, not a practical method. It does not claim that three matching tests are optimal; no example is claimed to show that two, or one, must fail; and the *and* between the matching tests cannot be replaced by an *or* on the strength of this article. It does not claim that a false-positive parameter exists: the bound $|\mathcal{B}_f| \leq 2$ is an upper bound, not a maximum, and no nonempty $\mathcal{B}_f$ is known. It neither improves Part I's 196-point simultaneous-separation bound nor contradicts its sharpness; it solves a weaker decision problem under a stronger irreducibility hypothesis, which is essential to the proof. The single triple test is safe but not complete. The implementation is an exact rational prototype, not a verified extraction; it rejects reducible input, does not implement the multiplicity shortcut, and the extension to reducible sextics (with a quintic criterion) is not included. The checks are finite exact tests, not formal proofs; the independent classifier shares SymPy's arithmetic library; forty-eight examples are not exhaustive, and none of them is a nonsolvable false positive. No benchmark against Magma, PARI/GP or another production package was made, and the recorded running time is an environment observation. The research questions are open within the article, not claimed to be unrecorded in the literature. The command-line parser is meant for trusted local input.

At the write, the manuscript's verifier was rerun on a copy with the delivered file names (Python 3.13.5, SymPy 1.14.0; the *Rerunning* section of the README gives the commands): it exited 0, its JSON output equals `data/02-three-plus-one-verification.json` after removing carriage returns except for `elapsed_seconds`, and its console output equals `data/02-three-plus-one-verification.log` except for the final *wrote* line, which names the output path. At placement a separate script (not shipped) recomputed the six pentads and their incidences, the pentad sums [\(23.3\)](#) symbolically, the 45 shared-edge factorizations and [\(20.4\)](#); numerically, at 60 digits and not as a proof, it found no invariant pentad for the A_6 seed $x^6 + 24x - 20$ and exactly one for the $\text{PSL}_2(5)$ and $\text{PGL}_2(5)$ seeds of the verifier, with no common zero of the pentad differences, consistent with [theorem 24.1](#) and with $J_f = 1$ there. None of this is an independent proof review.

A consequence noted at the write (not stated in either manuscript). Let f be an irreducible nonsolvable sextic over K and $t \in K$. By [theorem 22.2](#) the values $k_T(t)$ are distinct, so Part I's $L_f(t) \neq 0$ by [\(5.2\)](#); by [lemma 23.2](#) every $q_e(t) \neq 0$, so $Q_f(t) \neq 0$. Hence $E_f(t) = 0$ exactly when $U_f(t) = 0$: at base-field parameters such an f can fail to be separated only through a collision of two block-disjoint matchings. This bears on Research question [16.3](#), which it does not settle.

19 The question: must a decision algorithm separate everything?

The formal sextic project in ProveIt already proves a correct total-recursive radical-solvability decision for integer sextics. Its final decision file explicitly does not claim primitive recursiveness, because the construction uses unbounded search for separating invariant evaluations [\[5\]](#). A subsequent research report compresses the relevant partition descriptors and proves that one of $1, \dots, 196$ simultaneously separates all fifteen matching values and all ten triple-partition values [\[17\]](#). That report is not a formal theorem of the Lean project.

[write] The report cited here is Part I. The separator report [\[17\]](#) is Part I of this report, printed above; its text at the manuscript's pin is Part I without the dated notes of this write. Its 196-point bound is [theorem 9.1](#).

The present article asks a different question. Is full separation actually necessary for the *Boolean radical-solvability decision*? A collision between two nonrational invariant values may be harmless. Even a rational value with several antecedents may be harmless if its occurrence at a few independent parameters forces a genuine block system.

Our answer is that, for irreducible sextics, the stronger separation problem can be bypassed. The decision

only needs three fixed matching evaluations and one fixed triple evaluation. The important work is not the final root-counting step; it is the proof that every potentially misleading rational matching fiber must be the same five-element configuration.

19.1 A precise statement of the main theorem

Let K have characteristic zero and let

$$f(X) = X^6 - e_1X^5 + e_2X^4 - e_3X^3 + e_4X^2 - e_5X + e_6 \in K[X] \quad (19.1)$$

be irreducible. Section 20 defines monic polynomials $R_{2,f}(t, Z)$ and $R_{3,f}(t, Z)$ of Z -degrees 15 and 10, respectively. Write $\text{Root}_K(P)$ when P has a root in K .

Theorem 19.1 (Three plus one criterion). *For any distinct $t_1, t_2, t_3 \in K$ and any $u \in K$,*

$$f \text{ is solvable by radicals over } K \iff \text{Root}_K(R_{3,f}(u, Z)) \vee \bigwedge_{j=1}^3 \text{Root}_K(R_{2,f}(t_j, Z)). \quad (19.2)$$

In particular $t_1 = 1, t_2 = 2, t_3 = 3, u = 1$ works for every irreducible rational sextic. Repeated roots of the specialized resolvents are allowed.

The proof is completed in Section 25. The formula has an *and* between the three matching tests, and an *or* between the matching branch and the triple branch. Replacing the former by an *or* is not justified by this article.

19.2 What is classical, inherited, and proposed here

Solvability by radicals is equivalent to solvability of the Galois group in characteristic zero. The sextic block-system criterion using the two solvable wreath products is classical; Boswell and Glasser give sextic resolvent criteria in this setting [2]. More generally, Landau and Miller proved polynomial-time decidability of radical solvability in 1985 [1]. Modern computational Galois theory provides much broader splitting-field and radical-tower algorithms [14].

The descriptors $D_{\mathcal{B}}$ and their compression into $h_P(t)$ and $k_T(t)$ are inherited from the ProveIt formal project and its separator report [7, 17]. We reprove the algebraic facts needed here, so the new argument does not assume the correctness of every claim in an unrefereed predecessor.

The contributions developed in this article are the nonsolvable-case triple rigidity, the pentad classification of rational matching fibers, the explicit forced rational value and multiplicity, the two-point exceptional-set theorem, the three-plus-one decision criterion, and the displayed coefficient-only matrix realizations with a tested implementation. They are proposed contributions relative to the inspected source, not certified claims of first publication. In particular we do not infer novelty merely from the absence of a statement in the repository.

19.3 Scope and non-claims

Irreducibility is essential to the proof: it supplies transitivity and ensures that a base-field parameter is not a root of f . The matrix identities, in contrast, make sense for a general separable sextic, subject to the stated nonvanishing condition for the triple formula.

The new theorem neither improves the predecessor's 196-point *simultaneous-separation bound* nor contradicts its sharpness statement for arbitrary complex test sets. It solves a weaker decision problem under a stronger irreducibility hypothesis. We do not determine the optimal number of matching tests. In particular, no example in this package is claimed to show that two, or even one, must fail.

The repository was inspected at commit

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and was not modified or rebuilt. The new proofs are conventional mathematical proofs, not imported Lean theorems. The implementation is an exact rational prototype, not a verified extraction from the existing formal development.

20 Complete descriptors and the compressed resolvents

Let L be the splitting field of f over K , and label its distinct roots $\alpha_1, \dots, \alpha_6$. Let $G = \text{Gal}(L/K) \leq S_6$. Put $\mathcal{E} = \binom{[6]}{2}$. Let $\mathcal{M}$ be the fifteen perfect matchings of $[6]$, and let $\mathcal{T}$ be the ten partitions of $[6]$ into two unordered triples. The action of G on roots induces actions on all three sets.

For a block B , set

$$p_B(u) = \prod_{i \in B} (u - \alpha_i), \quad D_{\mathcal{B}}(u, v) = \prod_{B \in \mathcal{B}} (v - p_B(u)).$$

These are complete descriptions of the block partitions, before evaluation.

Lemma 20.1 (Completeness). *Distinct partitions in either $\mathcal{M}$ or $\mathcal{T}$ have distinct descriptors.*

Proof. The roots of p_B among the distinct α_i recover exactly the set B . Thus different blocks have different root polynomials. Regard $D_{\mathcal{B}}$ as a polynomial in v over $L(u)$. Its monic linear factors recover the set of polynomials p_B , and hence the blocks. Unique factorization in $L(u)[v]$ proves the assertion. $\square$

For a pair $b = \{i, j\}$, write $s_b = \alpha_i + \alpha_j$ and $p_b = \alpha_i \alpha_j$. For $P \in \mathcal{M}$, define

$$\begin{aligned} A_P &= \sum_{b \in P} p_b, & B_P &= \sum_{\substack{b, c \in P \\ b \neq c}} s_b p_c, & C_P &= \sum_{\{b, c\} \in \binom{P}{2}} p_b p_c, \\ h_P(t) &= A_P t^2 + B_P t - C_P. \end{aligned} \tag{20.1}$$

The sum in B_P is ordered: it contains six products $s_b p_c$, not three. For $T \in \mathcal{T}$, put

$$a_T = \sum_{B \in T} e_2(B), \quad b_T = \sum_{B \in T} e_3(B), \quad k_T(t) = a_T t - b_T. \tag{20.2}$$

Here $e_j(B)$ denotes the j th elementary symmetric function of the roots in B . Define

$$R_{2,f}(t, Z) = \prod_{P \in \mathcal{M}} (Z - h_P(t)), \quad R_{3,f}(t, Z) = \prod_{T \in \mathcal{T}} (Z - k_T(t)). \tag{20.3}$$

Their coefficients lie in $K[t]$ by Galois invariance. More strongly, their coefficients are integer polynomials in $t, e_1, \dots, e_6$, because the root expressions are symmetric integer polynomials.

Proposition 20.2 (Compression identities). *For every $P \in \mathcal{M}$ and $T \in \mathcal{T}$,*

$$D_P(u, v) = v^3 - (3u^2 - e_1u + A_P)v^2 + (3u^4 - 2e_1u^3 + (e_2 + A_P)u^2 - B_Pu + C_P)v - f(u), \quad (20.4)$$

$$D_T(u, v) = v^2 - (2u^3 - e_1u^2 + a_Tu - b_T)v + f(u). \quad (20.5)$$

Consequently $P \mapsto h_P$ is injective into $L[t]$, and $T \mapsto k_T$ is injective into $L[t]$.

Proof. The three pair-block polynomials are $u^2 - s_bu + p_b$. Their sum is $3u^2 - e_1u + A_P$. Their pairwise product sum is $3u^4 - 2e_1u^3 + (e_2 + A_P)u^2 - B_Pu + C_P$: for the coefficient of u^2 , use $e_2 = A_P + \sum_{b < c} s_b s_c$. Their total product is $f(u)$. Expanding the cubic in v proves (20.4). The sum of the two cubic block polynomials is $2u^3 - e_1u^2 + a_Tu - b_T$, and their product is $f(u)$, proving (20.5). Equality of two compressed polynomials gives equality of all the partition-dependent coefficients in the corresponding displayed descriptor. Lemma 20.1 then proves injectivity. $\square$

On the curve used by the predecessor report,

$$\begin{aligned} D_P(t, 2t^2) &= K_2(t) - 2t^2 h_P(t), & K_2(t) &= 2t^6 + 2e_2t^4 - f(t), \\ D_T(t, 2t^2) &= K_3(t) - 2t^2 k_T(t), & K_3(t) &= 4t^4 - 4t^5 + 2e_1t^4 + f(t). \end{aligned} \quad (20.6)$$

For $t \neq 0$, these are invertible affine changes of the specialized values. Thus root existence and multiplicity transfer to the original descriptor resolvents. The compressed polynomials also make sense at $t = 0$, whereas the original descriptors on this curve collapse there.

[write] Same objects as Part I. Lemma 20.1 is lemma 2.1; proposition 20.2 and (20.6) are propositions 3.1 and 3.2, whose displays (3.3) and (3.6) give the same K_2 and K_3 . The definitions (20.1) and (20.2) are (3.1) and (3.4). The resolvents (20.3) are Part I's (7.1) with the arguments in the other order: $R_{2,f}(t, Z)$ here is $R_{2,f}(Z, t)$ there.

Remark 20.3 (A specialized rational root need not label a single partition). If $c \in K$ is a root of a specialized resolvent, its fiber is a nonempty G -invariant set of partitions. Without injectivity, that fact alone does not make any one partition G -fixed. This distinction is the reason for the collision analysis below.

21 Two group lemmas on six vertices

We use no list of the sixteen transitive subgroups of S_6 in the proof. The first lemma is classical; the second reduces the relevant degree-six obstruction to very small graphs.

Lemma 21.1 (The block criterion). *A transitive subgroup $G \leq S_6$ is solvable if and only if it fixes an element of $\mathcal{M}$ or of $\mathcal{T}$.*

Proof. A group fixing a matching embeds in $S_2 \wr S_3$, and a group fixing a partition into two triples embeds in $S_3 \wr S_2$. Both wreath products are solvable, since their base groups and quotient groups are solvable. Subgroups of solvable groups are solvable.

Conversely, a finite solvable primitive permutation group has prime-power degree. Here is the needed argument. Choose a minimal nontrivial normal subgroup N . Its commutator subgroup is characteristic in N and normal in G ; minimality and solvability imply that N is abelian. For a prime p dividing $|N|$, the subgroup $N[p]$ is characteristic and nontrivial, so $N = N[p]$ is elementary abelian. The N -orbits form a block system. Primitivity and faithfulness therefore make N transitive. An abelian transitive permutation

group is regular: all point stabilizers in it agree, and their intersection is the kernel, which is trivial. The degree is then $|N| = p^r$. Since six is not a prime power, a solvable transitive $G \leq S_6$ is imprimitive. The possible sizes of its nontrivial blocks are two and three, so it fixes an element of $\mathcal{M}$ or $\mathcal{T}$. $\square$

[write] **Part I proves the same criterion.** Lemma 21.1 is lemma 10.1 together with the equivalence of solvability and a fixed pair or triple partition in proposition 10.2, proved there by the same minimal-normal-subgroup argument.

Lemma 21.2 (Invariant regular graphs). *If a transitive subgroup of S_6 preserves a simple regular graph of degree $r \in \{1, 2, 3, 4\}$, then the subgroup is solvable. Consequently every nonsolvable transitive subgroup of S_6 is transitive on $\mathcal{E}$.*

Proof. A 1-regular graph is a perfect matching; its automorphism group is $S_2 \wr S_3$. A 2-regular graph on six vertices is either a six-cycle or two disjoint triangles. Their automorphism groups are a dihedral group of order twelve and $S_3 \wr S_2$, respectively. All are solvable. Taking complements handles degrees four and three.

Now take an orbit of G on edges. Its union is a G -invariant graph, and vertex transitivity makes that graph regular. A nonempty proper edge orbit has degree between one and four: degree five would be the entire K_6 . The first part rules out a proper edge orbit for nonsolvable G . $\square$

The proof of Lemma 21.2 also shows why degree six is special. A proper invariant graph in a higher degree need not have a solvable automorphism group. The rest of the argument uses this small-degree rigidity rather than an enumeration of Galois groups.

22 Multiplicative rigidity and a safe triple test

Throughout this section assume that G is nonsolvable. For a base-field parameter $u \in K$, set $\beta_i = u - \alpha_i$. The β_i are distinct and nonzero: f is irreducible of degree six, hence has no root in K .

Lemma 22.1 (Distinct shifted pair products). *The fifteen products $\beta_i\beta_j$, $1 \leq i < j \leq 6$, are pairwise distinct.*

Proof. Equality of two pair products defines a G -invariant equivalence relation on $\mathcal{E}$. By the edge transitivity of Lemma 21.2, all equivalence classes have the same size m . Two distinct edges with equal products cannot share a vertex: cancellation of its nonzero β would equate two other β 's. Thus each class is a matching and $m \leq 3$. Also m divides $|\mathcal{E}| = 15$, so $m = 1$ or $m = 3$.

If $m = 3$, there are five classes, each a perfect matching. The common value c of a class satisfies

$$c^3 = \prod_{i=1}^6 \beta_i = f(u).$$

Different classes give five distinct values c , impossible for a cubic polynomial. Hence $m = 1$. $\square$

Theorem 22.2 (Triple rigidity at every base-field point). *If G is nonsolvable, then for every $u \in K$ the ten values $k_T(u)$ are pairwise distinct, and $R_{3,f}(u, Z)$ has no root in K .*

Proof. Any two distinct partitions into two triples can be relabeled

$$T = abc \mid def, \quad T' = abd \mid cef.$$

Indeed, after replacing a block by its complement when necessary, two chosen three-element blocks have intersection size two. The difference of the sums of their shifted block products is

$$\begin{aligned} & \beta_a\beta_b\beta_c + \beta_d\beta_e\beta_f - \beta_a\beta_b\beta_d - \beta_c\beta_e\beta_f \\ &= (\beta_c - \beta_d)(\beta_a\beta_b - \beta_e\beta_f), \end{aligned}$$

which is nonzero by distinctness of the roots and Lemma 22.1. By (20.5), the sum of the shifted block products is $2u^3 - e_1u^2 + k_T(u)$. Hence the $k_T(u)$ are distinct.

If one value were in K , its unique antecedent T would be fixed by every element of G , by equivariance and injectivity. Lemma 21.1 would then make G solvable, a contradiction. $\square$

Thus a single triple test is unconditionally safe as a *positive* solvability certificate for irreducible sextics. It is not a complete test: a solvable group can preserve three pairs without preserving two triples. Those groups require the matching branch.

23 Matching collisions, pentads, and rational fibers

23.1 Shared-edge collisions cannot occur in the nonsolvable case

For an edge $e = \{i, j\}$ define

$$q_e(t) = t^2 + (\alpha_i + \alpha_j)t - \alpha_i\alpha_j. \quad (23.1)$$

Lemma 23.1 (Shared-edge factorization). *If distinct matchings P, Q share an edge e , then*

$$h_P(t) - h_Q(t) = (A_P - A_Q)q_e(t), \quad A_P - A_Q \neq 0. \quad (23.2)$$

Proof. Let the four remaining roots be a, b, c, d . Changing their pairing changes its product sum among $ab + cd, ac + bd, ad + bc$. The differences are products of differences of distinct roots, for example $ab + cd - ac - bd = (a - d)(b - c)$, so $\Delta A \neq 0$.

The contribution of the two changing blocks to B is always the third elementary symmetric polynomial of a, b, c, d ; the common block contributes $s_e\Delta A$ to the change in B . The product of the two changing pair products is always $abcd$, so the change in C is $p_e\Delta A$. Therefore $\Delta h = \Delta A(t^2 + s_et - p_e)$. $\square$

[write] **Also in Part I.** This is lemma 4.1, with q_e as in (4.1), proved there by the same argument.

Lemma 23.2 (No shared edge in an equal-value fiber). *Suppose G is nonsolvable and $t \in K$. Then $q_e(t) \neq 0$ for every edge e . In particular, distinct matchings with equal h -values at t are edge-disjoint.*

Proof. The edges satisfying $q_e(t) = 0$ form a G -invariant graph. For a fixed vertex i , the equation is

$$(t - \alpha_i)\alpha_j = -t^2 - t\alpha_i.$$

Since $t \neq \alpha_i$, there is at most one possible partner j . The graph has maximum degree one. Vertex transitivity makes it either empty or a perfect matching. The latter is impossible by Lemma 21.1. The final assertion follows from Lemma 23.1. $\square$

23.2 The six pentads

Definition 23.3. A *pentad* is a set of five pairwise edge-disjoint perfect matchings of K_6 . Equivalently it is a one-factorization of K_6 .

Lemma 23.4 (The small incidence geometry). *There are exactly six pentads. Each matching belongs to exactly two pentads, and any two distinct pentads intersect in exactly one matching.*

Proof. The fifteen matchings and all six pentads are listed in Appendix C. Here is a completeness argument behind the table. Fix $P_0 = 12 \mid 34 \mid 56$. A pentad containing P_0 must contain exactly one matching using the edge 13. The matchings containing 13 and disjoint from P_0 are $13 \mid 25 \mid 46$ and $13 \mid 26 \mid 45$.

For either choice, the union of the two chosen matchings is a six-cycle, and its complement in K_6 is a triangular prism. That prism has exactly one one-factorization. To see this, call its three edges between the triangles vertical. A perfect matching contains either all three vertical edges, or one vertical edge together with the opposite edge in each triangle. The all-vertical matching cannot occur in a one-factorization, because the remaining two triangles have no perfect matching. The other three matchings must therefore all occur, and they do partition the prism's edges. Thus P_0 lies in exactly two pentads. Relabeling proves the same for every matching, and incidence counting gives $15 \cdot 2/5 = 6$ pentads.

Two pentads sharing two matchings would give two one-factorizations of the same complementary triangular prism, so they would coincide. Hence distinct pentads share at most one matching. The fifteen matchings, each belonging to two pentads, account for fifteen distinct unordered pairs of pentads; there are exactly $\binom{6}{2} = 15$ such pairs. Every pair therefore shares one. $\square$

23.3 Classification of the rational fibers

Theorem 23.5 (The pentad obstruction). *Assume G is nonsolvable and $t \in K$. If $c \in K$ is a root of $R_{2,f}(t, Z)$, then its fiber*

$$\Sigma = \{P \in \mathcal{M} : h_P(t) = c\}$$

is a G -invariant pentad. The root c has multiplicity exactly five. There is at most one root in K at this parameter. Moreover, G fixes at most one pentad, independently of the parameter.

Proof. The fiber is nonempty and G -invariant because $t, c \in K$. Its matchings are pairwise edge-disjoint by Lemma 23.2; let its size be r . Their union is a G -invariant r -regular graph, with $1 \leq r \leq 5$. Lemma 21.2 excludes $r = 1, 2, 3, 4$. Hence $r = 5$, giving a pentad and multiplicity exactly five in the product (20.3).

Two different rational roots would have disjoint fibers, whereas two pentads always meet. Thus there is at most one rational root. Finally, if G fixed two distinct pentads as sets, their singleton intersection would be a G -fixed matching. Lemma 21.1 again gives a contradiction. $\square$

A useful feature is that the proof excludes all possible fiber sizes without using orbit tables. In particular, the number five is forced by the geometry of K_6 , rather than guessed from an exceptional subgroup of S_6 .

Lemma 23.6 (Pentad averages). *For every pentad Σ ,*

$$\sum_{P \in \Sigma} A_P = e_2, \quad \sum_{P \in \Sigma} B_P = 3e_3, \quad \sum_{P \in \Sigma} C_P = e_4. \quad (23.3)$$

Proof. Every edge occurs exactly once in a pentad, proving the first identity. For a triple of roots $\alpha_i, \alpha_j, \alpha_k$, each of its three internal edges occurs in one matching of the pentad. In the corresponding B_P , the remaining

root lies in another block and contributes the triple monomial exactly once. This accounts for coefficient three in $3e_3$.

For a four-element set of roots, its complementary edge occurs in exactly one matching of the pentad. That matching pairs the four remaining roots in exactly one way, and contributes their product once to C_P . Conversely, every term of every C_P arises this way. This proves the third identity. $\square$

Corollary 23.7 (The forced value). *Under the nonsolvable hypothesis, every base-field root of $R_{2,f}(t, Z)$ is*

$$\mu_f(t) = \frac{e_2 t^2 + 3e_3 t - e_4}{5}. \quad (23.4)$$

Consequently any base-field root with another value, or with multiplicity other than five, certifies solvability of an irreducible sextic.

Proof. Sum $h_P(t) = c$ over the five members of its pentad fiber and apply Lemma 23.6. The certificate statement is the contrapositive of this calculation and Theorem 23.5. $\square$

24 The exceptional polynomial and the two-point theorem

Let

$$\mathcal{B}_f = \{t \in K : R_{2,f}(t, Z) \text{ has a root in } K\}$$

when G is nonsolvable. These are precisely the parameters at which the matching test alone gives a false positive.

Theorem 24.1 (A canonical exceptional polynomial). *If G has no invariant pentad, then $\mathcal{B}_f = \emptyset$. Otherwise let Σ be its unique invariant pentad and define*

$$J_f(t) = \text{monic gcd}_{P \in \Sigma} (h_P(t) - \mu_f(t)) \quad \text{in } L[t]. \quad (24.1)$$

Then $J_f \in K[t]$, $J_f \neq 0$, and $\deg J_f \leq 2$. Moreover,

$$\mathcal{B}_f = \{t \in K : J_f(t) = 0\}. \quad (24.2)$$

In particular $|\mathcal{B}_f| \leq 2$.

Proof. If a rational root occurs, Theorem 23.5 supplies an invariant pentad. This proves the first assertion. Now suppose Σ exists. Not all the polynomials $h_P - \mu_f$, $P \in \Sigma$, are zero, because the h_P are pairwise distinct polynomials by Proposition 20.2. Thus their monic gcd is defined and nonzero, and its degree is at most two.

Every $\sigma \in G$ permutes the family whose gcd is being taken and fixes μ_f . It follows that $\sigma(J_f)$ is the same monic gcd, hence equals J_f . Its coefficients belong to $L^G = K$.

If $t \in \mathcal{B}_f$, its fiber is the unique invariant pentad Σ , and Corollary 23.7 makes each $h_P(t) - \mu_f(t)$ zero. Therefore t is a common zero, hence a zero of the gcd. Conversely a zero of the gcd makes all five values equal to $\mu_f(t) \in K$, so the matching resolvent has a base-field root. This proves (24.2). A nonzero polynomial of degree at most two has at most two roots in a field. $\square$

There is also a shorter proof of the cardinality bound alone. Any two false-positive parameters use the same invariant pentad. Choose two members $P \neq Q$ of that pentad. Every false-positive parameter is a zero of the single nonzero polynomial $h_P - h_Q$, of degree at most two. The gcd formulation additionally identifies a canonical exceptional polynomial over the base field.

Remark 24.2 (Existence versus an upper bound). The theorem is valid if $J_f = 1$, if J_f has no root in K , or even if $\mathcal{B}_f$ is empty for every nonsolvable irreducible sextic. We have not proved that any nonempty $\mathcal{B}_f$ occurs over $\mathbb{Q}$, and the verification sample below contains no such example. Accordingly, the number two is a proved upper bound, not a proved maximum.

25 Decision certificates and the proof of three plus one

Proof of Theorem 19.1. If f is solvable by radicals, its transitive Galois group is solvable. By Lemma 21.1, it fixes a matching P or a triple partition T . In the first case $h_P \in K[t]$, so every matching test succeeds. In the second case $k_T \in K[t]$, so the chosen triple test succeeds.

Conversely, suppose G is nonsolvable. Theorem 22.2 rules out the triple branch at every $u \in K$. Theorem 24.1 allows at most two matching parameters with a base-field root. Three distinct matching parameters therefore cannot all succeed. Thus neither branch of (19.2) can hold. $\square$

25.1 Positive and negative certificates

There are several ways for exact computations to terminate early. A root of $R_{3,f}(u, Z)$ in K is a positive certificate. A root c of $R_{2,f}(t, Z)$ with $c \neq \mu_f(t)$, or with multiplicity different from five, is another positive certificate. Without that shortcut, roots at all three distinct matching parameters give a positive certificate. A root-free triple resolvent together with a root-free matching resolvent at just one of the three chosen parameters gives a negative certificate.

A negative certificate still needs an exact certificate that the two polynomials have no base-field roots; numerical root approximations do not supply that fact. Over $\mathbb{Q}$, rational polynomial factorization or a bounded integer-root search after an integral construction suffices. The delivered prototype uses exact rational root extraction and records the roots and multiplicities; it does not implement the multiplicity shortcut, to keep the baseline rule transparent.

25.2 A general orbit-collapse bound

The elementary principle behind the parameter count applies more widely. Its role is explanatory, not a claim that finite root counting is new.

Proposition 25.1 (Orbit budget). *Let L/K be finite Galois, with group G , and let X be a finite G -set with no fixed point. Suppose $x \mapsto p_x(t) \in L[t]$ is equivariant and injective, with $\deg p_x \leq d$. If X has r orbits, then*

$$\left| \left\{ t \in K : \prod_{x \in X} (Z - p_x(t)) \text{ has a root in } K \right\} \right| \leq dr.$$

Proof. A base-field value has a G -invariant nonempty fiber, which contains an entire orbit. In each orbit choose two different points x, y . If that orbit collapses to one value at t , then $(p_x - p_y)(t) = 0$. This nonzero polynomial has at most d roots. Taking the union over the r orbits gives the bound. $\square$

For the matching problem, the stronger argument identifies every possible rational fiber with the same pentad, rather than budgeting for every orbit. That is why its bound is two rather than a multiple of two. This suggests a useful research strategy: classify invariant equal-value fibers before counting pairwise collisions.

26 Coefficient-only resolvents from compound matrices

The decision theorem would be less useful if evaluating its resolvents required already knowing the roots or the splitting field. This section gives explicit rational matrix formulas using only the sextic coefficients. The formulas also avoid a search for an elementary-symmetric representation of a large invariant polynomial.

All matrices below are over K . Let C be the 6×6 companion matrix representing multiplication by X on $K[X]/(f)$ in the basis $1, X, \dots, X^5$. For a linear map A , write $A^{\wedge k}$ for its multiplicative compound on the k th exterior power; its entries are $k \times k$ minors of A . Write $A^{[k]}$ for its additive compound, defined by

$$A^{[k]}(v_1 \wedge \dots \wedge v_k) = \sum_{j=1}^k v_1 \wedge \dots \wedge Av_j \wedge \dots \wedge v_k.$$

If A has eigenvalues λ_i and is diagonalizable, the two compounds have eigenvalues $\prod_{i \in I} \lambda_i$ and $\sum_{i \in I} \lambda_i$, respectively, in the same exterior basis. This follows immediately by applying the displayed definitions to wedges of eigenvectors.

Since f is separable, C diagonalizes over L . Consequently

$$S = C^{[2]}, \quad P = C^{\wedge 2} \quad (26.1)$$

are commuting 15×15 matrices with joint eigenvalues (s_e, p_e) indexed by the edges $e \in \mathcal{E}$. Repetition of some sums or products causes no problem; a common eigenbasis still exists.

26.1 The quartic data attached to an edge

Let $I = I_{15}$ and define

$$\begin{aligned} r_1 &= e_1 I - S, \\ r_2 &= e_2 I - P - S r_1, \\ r_3 &= e_3 I - P r_1 - S r_2, \\ r_4 &= e_4 I - P r_2 - S r_3. \end{aligned} \quad (26.2)$$

On the eigenspace indexed by $e = \{i, j\}$, the scalar eigenvalue of r_k is the k th elementary symmetric function of the four roots outside e . Indeed, if those four roots have elementary coefficients ρ_k , then

$$\sum_{k=0}^6 e_k z^k = (1 + s_e z + p_e z^2) \sum_{k=0}^4 \rho_k z^k.$$

Equating coefficients gives exactly the recurrence (26.2).

For four roots a, b, c, d , the three pairing sums

$$ab + cd, \quad ac + bd, \quad ad + bc$$

are roots of the classical quartic cubic resolvent

$$z^3 - \rho_2 z^2 + (\rho_1 \rho_3 - 4\rho_4)z - (\rho_3^2 + \rho_1^2 \rho_4 - 4\rho_2 \rho_4). \quad (26.3)$$

This identity can be proved by multiplying its three linear factors and collecting elementary symmetric terms; it is also checked symbolically by the supplied script. Define the commuting matrices

$$U = r_1 r_3 - 4r_4, \quad V = r_3^2 + r_1^2 r_4 - 4r_2 r_4. \quad (26.4)$$

Here U, V are matrices, not the descriptor variables u, v .

26.2 A 45-dimensional matching realization

For the chosen parameter t , put

$$Q = t^2 I + tS - P, \quad H = t^2 P + t(r_3 + Pr_1) - r_4. \quad (26.5)$$

For a matching containing an edge e , let z be the pairing sum of its remaining four roots. A direct use of (20.1) gives

$$h_P(t) = H_e + Q_e z. \quad (26.6)$$

In fact $A = p_e + z$, $B = s_e z + p_e \rho_1 + \rho_3$, and $C = p_e z + \rho_4$. Substituting these three identities proves (26.6).

Define a monic cubic matrix polynomial

$$\begin{aligned} N_t(Z) &= (ZI - H)^3 - r_2 Q(ZI - H)^2 + UQ^2(ZI - H) - VQ^3 \\ &= Z^3 I + N_2 Z^2 + N_1 Z + N_0, \\ N_2 &= -3H - r_2 Q, \\ N_1 &= 3H^2 + 2r_2 QH + UQ^2, \\ N_0 &= -H^3 - r_2 QH^2 - UQ^2 H - VQ^3. \end{aligned} \quad (26.7)$$

All products commute, since every matrix is a polynomial expression in the commuting S, P . Set

$$B_2(t) = \begin{pmatrix} 0 & 0 & -N_0 \\ I & 0 & -N_1 \\ 0 & I & -N_2 \end{pmatrix} \in \text{Mat}_{45}(K). \quad (26.8)$$

Theorem 26.1 (Matching characteristic-polynomial identity). *For every separable monic sextic and every $t \in K$,*

$$\det(ZI_{45} - B_2(t)) = R_{2,f}(t, Z)^3. \quad (26.9)$$

No nonvanishing assumption on any $q_e(t)$ is required.

Proof. Use the common eigenbasis of S, P over L . In that basis the three block coefficients of $B_2(t)$ are diagonal. Reordering the 45 basis vectors therefore decomposes $B_2(t)$ into fifteen scalar cubic companion matrices, one for each edge e . Their characteristic polynomials are

$$(Z - H_e)^3 - \rho_2 Q_e (Z - H_e)^2 + (\rho_1 \rho_3 - 4\rho_4) Q_e^2 (Z - H_e) - (\rho_3^2 + \rho_1^2 \rho_4 - 4\rho_2 \rho_4) Q_e^3.$$

By (26.3), this is the product of the three factors $Z - H_e - Q_e z$, over the three pairings of the remaining four roots. The identity is polynomial in Q_e ; in particular it remains true when $Q_e = 0$, when all three factors coincide.

By (26.6), these factors are precisely $Z - h_P(t)$ for matchings containing e . Each matching contains three edges, so the product over all fifteen edges counts each of the fifteen matching factors exactly three times. This proves (26.9) over L , hence over K . $\square$

The absence of division by Q_e is important. A construction that first solves $z = (Z - H_e)/Q_e$ would require special handling at shared-edge collisions. The homogeneous cubic (26.7) includes those collisions automatically.

26.3 A 20-dimensional triple realization

For $u \in K$ with $F = f(u) \neq 0$, set

$$D = (uI_6 - C)^{\wedge 3}, \quad B_3(u) = D + FD^{-1} - (2u^3 - e_1u^2)I_{20}. \quad (26.10)$$

The inverse exists because $uI_6 - C$ is invertible. This condition is automatic for irreducible f of degree six and $u \in K$.

Theorem 26.2 (Triple characteristic-polynomial identity). *For every separable monic sextic with $f(u) \neq 0$,*

$$\det(ZI_{20} - B_3(u)) = R_{3,f}(u, Z)^2. \quad (26.11)$$

Proof. On a wedge eigenvector indexed by a triple B , the eigenvalue of D is $z_B = \prod_{i \in B} (u - \alpha_i)$. The eigenvalue of FD^{-1} is $F/z_B = z_{B^c}$. The resulting eigenvalue of $B_3(u)$ is

$$z_B + z_{B^c} - (2u^3 - e_1u^2) = k_{\{B, B^c\}}(u).$$

Each unordered triple partition is counted twice, once for each complementary triple, proving (26.11). $\square$

Although D^{-1} is computed over the rational field in the prototype, the resulting resolvent has integer coefficients for a monic integral f and integral u . That integrality follows from the symmetric root formula, not from a numerical cancellation assumption.

26.4 Exact perfect-power extraction without factoring

The characteristic polynomials have degrees 45 and 20, but the desired resolvents have degrees 15 and 10. Their monic cube and square roots can be extracted by a triangular coefficient recurrence.

Suppose $A(Z)$ is monic of degree md and known to be $Q(Z)^m$, with

$$Q(Z) = Z^d + q_1Z^{d-1} + \cdots + q_d.$$

At degree $md - k$, the coefficient of Q^m is

$$mq_k + F_k(q_1, \dots, q_{k-1}), \quad 1 \leq k \leq d.$$

Exactly one factor can contribute q_k to that coefficient, and there are m choices of factor; every other contribution uses earlier coefficients. Thus q_k is determined by division by m . In characteristic zero this recovers the unique monic power root. The implementation then checks the full identity $Q^m = A$ exactly and raises an error if it fails.

This extra check guards the implementation against a wrong matrix formula or an indexing error. It is not a substitute for the mathematical proof of Theorems 26.1 and 26.2.

27 An explicit bounded arithmetic architecture

27.1 The delivered decision function

Over $\mathbb{Q}$ the algorithm takes a polynomial, makes it monic, verifies that it is an irreducible sextic, and proceeds as follows.

```

Build the companion matrix and the two compound matrices.
Compute R3(1, Z) by the 20-dimensional characteristic-polynomial identity.
If R3(1, Z) has a rational root, return SOLVABLE.
For t in (1, 2, 3):
    Compute R2(t, Z) by the 45-dimensional identity.
    If R2(t, Z) has no rational root, return NONSOLVABLE.
Return SOLVABLE.

```

Every characteristic-polynomial power identity is checked during extraction. The function records the rational roots and multiplicities used in its conclusion. The separate verification program invokes an independent Galois classifier, but the decision implementation itself never does.

The test values are fixed independently of the coefficient heights. This is a bounded number of resolvent evaluations, not a claim that each evaluation has unit cost or that factorization is free.

27.2 Root bounds and a primitive-recursive realization

For completeness one can avoid an abstract root-finding oracle entirely. Suppose first that f is monic integral, and let H be the maximum absolute value of its six nonleading coefficients. Put $R = 1 + H$. Every complex root satisfies $|\alpha_i| \leq R$: if $|z| > 1 + H$, then

$$H \sum_{j=0}^5 |z|^j < |z|^6,$$

so the leading term of $f(z)$ cannot be canceled by the others. From their definitions,

$$\begin{aligned} |A_P| &\leq 3R^2, & |B_P| &\leq 12R^3, & |C_P| &\leq 3R^4, \\ |a_T| &\leq 6R^2, & |b_T| &\leq 2R^3. \end{aligned}$$

Consequently, for $t = 1, 2, 3$,

$$|h_P(t)| \leq 27R^2 + 36R^3 + 3R^4 \leq 66R^4, \quad |k_T(1)| \leq 6R^2 + 2R^3 \leq 8R^3. \quad (27.1)$$

Both specialized resolvents are monic integer polynomials. Their rational roots must therefore be integers. Exact evaluation on

$$[-66R^4, 66R^4] \cap \mathbb{Z} \quad \text{and} \quad [-8R^3, 8R^3] \cap \mathbb{Z} \quad (27.2)$$

decides the respective root-existence predicates. This is a coarse termination bound, not a recommended practical root-finding method.

Proposition 27.1 (Bounded realization on the irreducible branch). *The three-plus-one criterion for monic irreducible integer sextics has a primitive-recursive realization using bounded integer and rational arithmetic. Irreducibility of an integral input can also be checked by a bounded search.*

Proof. The matrices have fixed dimensions. Their entries are obtained by a fixed number of rational operations; the triple inverse can be evaluated by the adjugate formula, whose denominator is nonzero on this branch. Determinants, characteristic polynomials, and the power-root recurrence are finite coefficient algorithms in fixed dimensions. Integer encoding of rational arithmetic, including reduction by the Euclidean algorithm or bounded gcd search, has a primitive-recursive realization. The final searches are bounded by (27.2).

For the last assertion, a proper monic factor of a monic integral sextic has integer coefficients, and its roots are a subset of the six roots. Each of its coefficients is bounded in absolute value by $\binom{d}{j}R^j$, hence by $64R^6$, for a factor of degree $d \leq 5$. Enumerate the finitely many monic polynomials of degrees one through five in that coefficient box and test exact divisibility. A proper factor exists exactly when one test succeeds. All bounds are explicit integer expressions in the input coefficients. $\square$

For an integral sextic with leading coefficient $a \neq 0$, monicization

$$F(Y) = a^5 f(Y/a)$$

produces a monic integral sextic and scales every root by a . It preserves the splitting field and radical solvability. Rational input coefficients can first be cleared of denominators. The actual prototype instead works directly over $\mathbb{Q}$ and normalizes the leading coefficient; it uses exact rational factorization rather than the deliberately coarse boxes.

27.3 Why this is not a new general decidability theorem

Landau–Miller already gives a stronger general complexity theorem [1]. Proposition 27.1 isolates a direct, bounded arithmetic architecture for these particular ProveIt descriptors. It does not claim a new abstract decidability result, a polynomial bit-complexity bound for the supplied code, or a speed advantage over production Galois algorithms.

The prototype rejects reducible input. A full sextic solver can factor first and handle the irreducible factors; degrees at most four are classically solvable, while a degree-five factor needs a quintic criterion. That extension is not silently included in the advertised three-plus-one theorem.

28 Exact verification and reproducibility

The package contains a coefficient-only implementation `code/sextic_resolvents.py`, a separate verifier `code/verify.py`, and recorded outputs in `data/verification.json` and `data/verification.log`. The recorded run used Python 3.13.5 and SymPy 1.14.0. The precise environment is also recorded in the JSON file. The following are finite exact tests, not formal proofs of the theorems.

[write] Shipped names. In this report the files named above are `code/02-three-plus-one-sextic_resolvents.py`, `code/02-three-plus-one-verify.py`, `data/02-three-plus-one-verification.json` and `data/02-three-plus-one-verification.log`; the delivered `requirements.txt` is `data/02-three-plus-one-requirements.txt`. The files `code/verify.py`, `data/verification.json` and `data/requirements.txt` of this directory are Part I's. The verifier imports `sextic_resolvents` under its delivered name and always writes `data/verification.json` beside the `code/` directory containing it, so run in the report directory under restored names it would overwrite Part I's recorded output. Run it only on a copy (section D).

28.1 Symbolic and split-root checks

The verifier enumerates all fifteen matchings, ten triple partitions, and six pentads. It checks the fifteen pentad intersections and the two-pentad membership of every matching. With six independent root variables it verifies all forty-five shared-edge collision identities, all eighteen pentad-sum identities, all twenty-five descriptor expansions, and the quartic cubic identity (26.3).

For ten explicit or deterministically generated rational root tuples, it compares the coefficient-only matrix answer with the direct root-product answer. There are thirty matching identities and thirty triple identities. For

split examples where $f(t) = 0$, the triple identity uses a recorded nearby rational parameter with $f(u) \neq 0$, as its hypothesis requires. Six of the thirty matching tests have repeated resolvent roots. Thus the comparison also exercises the collision cases; no squarefreeness filter is imposed.

28.2 All sixteen transitive group types

The sixteen seed sextics are the examples in SymPy 1.14.0's public Galois-group test source, attributed there to Cohen [16, 15]. For each seed we also test $f(X + 2)$ and $2^6 f(X/2)$, giving forty-eight irreducible inputs. An exact independent Galois-group computation checks the stated group type; its permutation group's solvability is then compared with the new criterion. The proof of the criterion itself does not use this classification.

All forty-eight comparisons agree. The public decision API takes the triple branch on twenty-two examples, the three-matching branch on fourteen, and the negative branch on twelve. All four nonsolvable group types occur, each in three coordinate variants. For every nonsolvable example the triple resolvent is squarefree and root-free over $\mathbb{Q}$. None of these examples has a rational matching root at 1, 2, 3, so the sample does not test an actual nonsolvable false positive.

Table 4 records the seed cases. An entry in a root column is a rational root, not the full factorization; braces indicate several roots. All roots displayed in this seed table have multiplicity one. The group names are the labels used by the cited implementation; S_4^+ and S_4^- denote its two degree-six embeddings, and $G_{36}^\pm$ its corresponding order-36 labels.

Group	Order	Solvable	$R_3(1)$ roots	$R_2(1)$ roots	$R_2(2)$ roots	$R_2(3)$ roots
C_6	6	yes	-3	-2	5	18
S_3	6	yes	0	$\{0, \pm 54\}$	$\{0, \pm 108\}$	$\{0, \pm 162\}$
D_6	12	yes	0	0	0	0
A_4	12	yes	$\emptyset$	3	3	3
G_{18}	18	yes	3	$\emptyset$	$\emptyset$	$\emptyset$
$A_4 \times C_2$	24	yes	$\emptyset$	3	3	3
S_4^+	24	yes	$\emptyset$	4	4	4
S_4^-	24	yes	$\emptyset$	10	25	46
G_{36}^-	36	yes	2	$\emptyset$	$\emptyset$	$\emptyset$
$S_4 \times C_2$	48	yes	$\emptyset$	-2	-2	-2
$\text{PSL}_2(5)$	60	no	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
$\text{PGL}_2(5)$	120	no	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
G_{36}^+	36	yes	8	$\emptyset$	$\emptyset$	$\emptyset$
G_{72}	72	yes	4	$\emptyset$	$\emptyset$	$\emptyset$
A_6	360	no	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
S_6	720	no	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

Table 4: Exact rational roots in the sixteen seed cases. The complete input polynomials and all forty-eight results are in the verification JSON.

Six malformed-input tests also pass: degree mismatch, reducibility, zero polynomial, repeated test parameters, too few parameters, and a zero parameter under the prototype's nonzero-parameter interface. The mathematical compressed criterion permits zero; the interface uses nonzero parameters to match the original descriptor convention without additional cases.

28.3 Two transparent examples

For $f(X) = X^6 - 3X^2 - 1$, the exact outputs are

$$\text{Roots}_{\mathbb{Q}} R_{3,f}(1, Z) = \emptyset, \quad \text{Roots}_{\mathbb{Q}} R_{2,f}(t, Z) = \{3\}, \quad t = 1, 2, 3.$$

The matching branch therefore certifies radical solvability even though the triple branch fails. In fact the displayed matching roots have multiplicity one, so Corollary 23.7 would already certify it at one parameter.

For $f(X) = X^6 + X + 1$, the coefficient-only triple formula yields

$$\begin{aligned} R_{3,f}(1, Z) = & Z^{10} - 24Z^8 + 66Z^7 + 195Z^6 - 1136Z^5 - 489Z^4 \\ & + 5503Z^3 + 2435Z^2 - 6361Z + 666. \end{aligned}$$

It has no rational root, and neither does $R_{2,f}(1, Z)$. The algorithm therefore returns nonsolvable without evaluating the remaining two matching resolvents. The supplied exact root routine, rather than numerical inspection of this displayed polynomial, establishes the root-free assertions.

28.4 What the checks do and do not establish

The independent classifier comparison checks the implementation against a separate computational route. It still shares the SymPy arithmetic library, and forty-eight examples are not an exhaustive proof. Likewise, symbolic identities test formulas, not every group-theoretic hypothesis in an infinite family. The logical justification is the written chain of lemmas above.

The implementation is a usable coefficient-only prototype rather than the root-level demonstration accompanying the predecessor separator report. No benchmark comparison with Magma, PARI/GP, or another production Galois package was performed. The package's running-time record is an environment observation, not a performance theorem.

29 Repository integration and proof obligations

The existing formal chain and this article meet at several precise places. The following source map concerns the inspected commit, not an assertion that any proposed module already exists.

Existing source	Role and boundary
<code>SexticSeparatingInvariants.lean</code>	Defines complete descriptors and proves injectivity and existence of separating evaluations. The compression identities here connect to those objects, but the new decision proof does not require a separating evaluation.
<code>SexticSeparatingSearch.lean</code>	Chooses separating parameters using unbounded minimization. The proposed new irreducible branch would bypass this selection, not prove that its existing selector returns one of 1, 2, 3.
<code>SexticSparseSymmetricSearch.lean</code>	Searches for elementary-symmetric certificates. The matrix formulas provide an alternative coefficient evaluator whose identities still require formalization.
<code>SexticReducibleDecision.lean</code>	Handles reducible sextics and includes its own search machinery. Those branches are not replaced merely by proving the present irreducible theorem.

Existing source	Role and boundary
<code>SexticRadicalDecision.lean</code>	Assembles the total-recursive decision and its semantic correctness theorem. The inspected file does not claim primitive recursiveness. A formalized bounded new branch would be an additional result, not an interpretation of an existing declaration.

All these files live under `Algebra/PolynomialFormulas/Lean/PolynomialFormulas/`. The source and predecessor report document these boundaries [5, 7, 11, 12, 10, 17].

A sensible formalization order begins with finite combinatorics. Encode the fifteen matchings and six pentads and prove the incidence statements, either structurally or by reflected finite computation. Next prove the invariant regular-graph lemma and its consequences for nonsolvable transitive actions. Then formalize the rational-fiber and exceptional-parameter arguments using Galois invariance and polynomial root bounds. Only after that is the exact matrix implementation needed for a fully coefficient-level decision.

The matrix path has separate obligations: the companion/exterior-power spectral identities, the quartic resolvent identity, exact characteristic polynomial powering, and the power-root recurrence. A verification-friendly alternative is to prove the matrix identities first for independent indeterminates, then specialize. Either route must distinguish formal identities from tested numerical instances.

To interface with the existing bivariate descriptor, Lean's coordinate order is $x(0) = v$ and $x(1) = u$, so the curve is represented by $x = (2t^2, t)$, not $(t, 2t^2)$ [7, 17]. The fixed points 1, 2, 3 therefore correspond to three fixed encoded parameters for the matching *decision tests*. They are not guaranteed separators of the full value set.

30 Further research questions

The following questions are deliberately narrower than a general request for better sextic algorithms. Each is tied to a specific remaining obstruction or implementation boundary. They are open within this article; no claim is made that each is an unrecorded open problem in the literature.

Question 30.1 (Can the exceptional set ever be nonempty?). Does there exist an irreducible nonsolvable $f \in \mathbb{Q}[X]$ and $t \in \mathbb{Q}$ such that $R_{2,f}(t, Z)$ has a rational root? If yes, produce an exact example with its multiplicity-five factor and nonsolvable Galois certificate. If no, prove that the pentad collapse equations have no admissible rational Galois specialization. The present tests do not decide this question.

Theorem 23.5 turns this from an unrestricted resolvent search into a constrained problem: the five matching values in one pentad must equal $\mu_f(t)$, while the original sextic remains irreducible and nonsolvable.

Question 30.2 (How many matching tests are really necessary?). Can three matching evaluations in Theorem 19.1 be reduced to two, or to one? There are distinct versions: a universal choice of one or two rational parameters; every choice of distinct parameters; and an adaptive choice depending on the sextic coefficients. The two-point upper bound does not prove optimality in any of these versions.

Question 30.3 (Classify the exceptional polynomial). For the unique invariant pentad, which degrees and factorization types can J_f have under the nonsolvable hypothesis? Determine whether degree two can occur, whether its discriminant is constrained by f , and whether a linear factor forces an additional block system. An answer could strengthen the criterion without changing the resolvent evaluators.

Question 30.4 (Eliminate the pentad from the coefficient description). Can J_f , or just its rational zero set, be recovered directly from $R_{2,f}(t, Z)$ and $\mu_f(t)$ using low-degree subresultants, without constructing a splitting field or choosing a pentad? A necessary and, in the nonsolvable case, sufficient condition is that $(Z - \mu_f(t))^5$ divide the specialized matching resolvent. The challenge is to distinguish that exceptional condition from genuine solvability and to keep coefficient growth controlled.

Question 30.5 (Extract useful witnesses, not only a Boolean answer). From a successful three-test certificate, can one recover a block system, a low-degree subfield, or the initial data of a radical tower efficiently? The decision proof does not identify which matching is fixed, and a rational root alone need not distinguish it from its colliding companions. A witness-producing refinement would connect more directly to radical expression construction.

Question 30.6 (Smaller coefficient-only realizations). Can the degree-15 and degree-10 resolvents be computed by rational matrices of those dimensions, or by structured determinant identities that avoid the 45- and 20-dimensional auxiliary characteristic polynomials? The present construction intentionally counts each matching three times and each triple partition twice. Removing that redundant incidence information without a coefficient-table search is a concrete algorithmic target.

Question 30.7 (Certified modular acceleration). Develop modular characteristic-polynomial evaluation, rational reconstruction, and independent certificates for the two exact power identities. Determine usable coefficient bounds and compare bit complexity and actual running time with standard Galois-group algorithms. The fixed matrix dimensions make this plausible, but the present article does not supply such a comparison.

Question 30.8 (Formalize the collision-aware criterion). Formalize Theorems 22.2, 23.5, 24.1, and 19.1 in Lean against the existing descriptor definitions. Which route is shortest and most reusable: abstract finite-group action arguments, a verified classification of the tiny regular graphs, or reflection over the explicit pentad table? Then verify the coefficient-only evaluator and connect the result to the semantic predicate.

Question 30.9 (Higher-degree collision geometry). For block resolvents of degrees eight, nine, or ten, which invariant fiber configurations can mimic a rational fixed point? Seek analogues of the unique pentad mechanism, or prove that several different invariant configurations force a larger parameter budget. Proposition 25.1 gives a baseline, but not the exceptional rigidity needed for a small bound.

Question 30.10 (Other characteristics and base fields). Which parts of the fiber classification survive for separable sextics in positive characteristic? The graph and cancellation arguments are largely characteristic-free, but averaging by five, power extraction by two or three, and the relation between radical towers and separability require separate treatment at small primes. Over characteristic-zero fields other than $\mathbb{Q}$, identify practical root-existence procedures compatible with the same three-plus-one theorem.

31 Conclusion

The central gain is a change in what has to be certified. To decide radical solvability of an irreducible sextic, it is unnecessary to prove that every specialized invariant value is distinct. In the nonsolvable case, the triple values separate automatically at base-field points. The only possible rational matching fiber is a pentad, its value is forced, its multiplicity is five, and the invariant pentad is unique. Polynomial injectivity then bounds all false-positive parameters by two.

This yields a universal three-plus-one decision rule and an exact, coefficient-only implementation. It leaves meaningful questions about the existence and optimality of the exceptional set, rather than assuming that the root-counting bound is sharp. The proof and implementation together provide a concrete new branch to formalize in ProveIt, without claiming that the repository already contains it or that classical decidability has been rediscovered as a new theorem.

C The matching and pentad tables

Label the matchings as follows. Each expression such as 12|34|56 denotes three unordered pairs, and the order of the pairs has no significance.

1	12 34 56	6	13 26 45	11	15 24 36
2	12 35 46	7	14 23 56	12	15 26 34
3	12 36 45	8	14 25 36	13	16 23 45
4	13 24 56	9	14 26 35	14	16 24 35
5	13 25 46	10	15 23 46	15	16 25 34

The six pentads, written as sets of matching labels, are

$$\begin{aligned}\Sigma_1 &= \{1, 5, 9, 11, 13\}, & \Sigma_2 &= \{1, 6, 8, 10, 14\}, \\ \Sigma_3 &= \{2, 4, 8, 12, 13\}, & \Sigma_4 &= \{2, 6, 7, 11, 15\}, \\ \Sigma_5 &= \{3, 4, 9, 10, 15\}, & \Sigma_6 &= \{3, 5, 7, 12, 14\}.\end{aligned}$$

The proof in Lemma 23.4 establishes completeness, independently of the code enumeration. Checking the displayed sets verifies each individual incidence used there. The Python representation is zero-based, so its stored matching indices are these labels minus one.

D Rebuilding and running the package

The article uses an internal bibliography and standard TeX packages. From the package root, run

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
python -m pip install -r requirements.txt
python code/verify.py
python code/sextic_resolvents.py 'x**6-3*x**2-1'
python code/sextic_resolvents.py 'x**6+x+1'
```

The shell build script runs LaTeX enough times to stabilize references. The Python verifier writes fresh verification JSON and a console report; redirect the latter to regenerate the recorded log. The generated split-root tuples use a fixed seed, and all mathematical arithmetic in the checks is exact.

For a Python program, import the module from code/ and call

```
import sympy as sp
from sextic_resolvents import decide_irreducible_sextic
x = sp.Symbol('x')
result = decide_irreducible_sextic(x**6 - 3*x**2 - 1, x)
assert result['solvable'] is True
print(result['evidence'])
```

The command-line expression parser is intended for trusted local expressions, not untrusted network input. The mathematical API takes SymPy polynomial objects or expressions and rejects nonsextic and reducible inputs.

[write] The shipped layout. The commands above assume the delivered package; do not run them in the report directory. code/02-three-plus-one-build.sh changes to its own directory, code/, which has no article.tex

(and the source it built is not shipped); build the report with `latexmk-pdf` from a copy of the directory. The evaluator imports nothing local and writes nothing, so it can be run in place under its shipped name:

```
uv run --no-project --with sympy==1.14.0 python code/02-three-plus-one-sextic_resolvents.py 'x**6+x+1'
```

The verifier must run on a copy with the delivered names, from an empty scratch directory `W`:

```
mkdir -p W/code W/data
cp code/02-three-plus-one-verify.py W/code/verify.py
cp code/02-three-plus-one-sextic_resolvents.py W/code/sextic_resolvents.py
cd W && uv run --no-project --with sympy==1.14.0 python code/verify.py
```

Then compare `W/data/verification.json` with `data/02-three-plus-one-verification.json` after removing carriage returns (on Windows the script writes CRLF); only `elapsed_seconds` differs.

E Proof dependency and audit checklist

The main logical chain is

small regular graphs $\implies$ edge transitivity $\implies$ shifted-product rigidity $\implies$ safe triple test,

and, separately,

shared-edge factorization $\implies$ disjoint matching fibers $\implies$ unique pentad $\implies$ at most two bad parameters.

The classical block criterion joins those two chains into Theorem 19.1. The coefficient-only matrix theorems then implement the already-proved resolvents; they do not enter the group-theoretic argument.

An independent reviewer should particularly check the use of irreducibility at every cancellation, the classification of 2-regular graphs on six vertices, completeness of the pentad list, the uniqueness of a G -invariant pentad, and the distinction between equality of specialized values and equality of their underlying polynomials. For the code, the exterior-basis sign convention, the signs in N_0 , and exact square/cube-root reconstruction are the most sensitive points. All receive either explicit proofs or separate identity checks above; none is justified only by floating-point evidence.

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